AI Playbook and Toolkit for Public Health Departments

Website Complete Reference Guide

This PDF compiles public website content from the Start Here, Learn, AI Support Areas, Assess, Maturity Model, Plays, Toolkit, Case Studies, Resources, and In the News pages. My Account, Organization Hub, Community, and Contribute areas are intentionally excluded from this document.

Contents

Executive Overview

The AI Playbook and Toolkit helps public health departments adopt generative and agentic AI responsibly through phased planning, governance, readiness assessment, stakeholder engagement, workforce development, use case prioritization, funding, deployment, oversight, and evaluation.

The ASPPH AI framework report has been incorporated through foundations on AI as a systems shift, digital determinants of health, human-centered AI, staff AI literacy, automation bias, missing voices analysis, transparency, tiered data use, environmental impacts, and the bridge professional role.

Implementation Phases

Phase	Range	Summary
PLAN	Plays 1-7	Vision, readiness, governance, stakeholders, workforce, change planning, use cases
BUILD	Plays 8-9	Funding and roadmap
DEPLOY	Plays 10-11	Change execution, build, deploy
GOVERN	Plays 12-13	Monitor, evaluate, oversight

Background Foundations

Staff Training and Shared Readiness

Use these modules to build your own foundation before you participate in AI project work. The learning section can support self-paced learning, onboarding, leadership orientation, governance preparation, and role-specific readiness.

This overview explains how you can use the learning section as a staff training curriculum, not just a reference library. You can use the modules before workshops, during facilitated sessions, or as onboarding if you will participate in AI-supported work.

The goal is to help you understand your role in shared readiness. You may need to understand what AI can and cannot do in their workflows. If you are in a leadership role, you need to understand governance and resource implications. If you work in privacy, legal, procurement, IT, communications, equity, or data, you need enough shared language to review proposals consistently.

You can adapt the modules into short training sessions by pairing the narrative content with the discussion questions and staff exercise. The expected takeaways should become practical implementation artifacts, such as a training plan, a list of governance questions, or a completed readiness input.

Module Overview

These learning modules are designed to help health departments create a shared baseline before you begin AI project work. They can be used for staff onboarding, leadership orientation, governance preparation, cross-program training, or pre-work before readiness assessment and use case prioritization.

How You Can Use the Learning Modules

- Use the modules as self-paced reading before workshops or governance meetings.
- Work through one module at a time as a 30- to 60-minute learning session.
- Complete modules as pre-work before completing the readiness assessment or prioritizing AI use cases.
- Use module exercises to generate artifacts that feed directly into playbook tools, governance review, and implementation planning.

Recommended Training Sequence

- Start with Understanding Artificial Intelligence, Generative AI, and Agentic AI so you have the basic vocabulary.
- Move into Risks, Human-Centered AI, Missing Voices, Public Transparency, and Tiered Data Use before use cases are selected.
- Use AI Literacy, Bridge Professional Role, Funding, procurement, privacy, validation, incident response, and role-based guidance when preparing implementation work.
- Use AI Support Areas after foundational topics so you can discuss practical opportunities without skipping governance.

Facilitation Tips

- Start each session with a concrete public health workflow rather than a general technology discussion.
- Ask yourself to identify what AI may support, what humans must decide, and what data safeguards are needed.
- End each session with a documented takeaway: a question for governance, a possible tool to complete, or a readiness gap to address.
- Keep examples local to the department's programs, authorities, data systems, and community context.

Why This Matters for Your Practice

Health departments can use the learning modules as practical training material rather than static background reading. The modules help staff, leaders, governance members, and program teams build a common foundation before you begin AI project work.

Reflection Questions

- Which staff groups need baseline AI training before tools are approved?
- Which modules should be required before governance participation, vendor review, or use case prioritization?
- How will the department document that staff understand data safeguards, human review expectations, and escalation pathways?

Practical Exercise

Create a short training plan that assigns modules to staff groups, identifies which exercises should be completed, and connects each module to a play or tool that the department will use next.

Expected Takeaways

- Staff AI training plan
- Module assignment matrix by role
- Training completion expectations
- List of governance or implementation artifacts produced during training

Understanding Artificial Intelligence

AI is a family of methods that enable systems to learn from data, recognize patterns, generate content, and support action toward defined goals.

Artificial intelligence should be introduced as a set of capabilities rather than a single product. In public health, AI may appear as predictive analytics, classification, natural-language summarization, generative drafting, document search, translation support, or workflow automation. Staff need to recognize these differences because each capability has different risks, data needs, and review requirements.

A shared foundation also helps departments avoid technology-first decisions. The central question is not whether an AI tool is impressive, but whether it fits a public health purpose, uses appropriate data, improves a workflow, and remains accountable to humans. This matters when AI is embedded in vendor products, offered through enterprise software, or used informally by staff.

By the end of this module, you should be able to describe an AI-supported workflow in plain language: what the system does, what data it uses, who reviews the output, what decision remains human, and what safeguards are required before use.

Module Overview

You do not need to become a machine learning engineer, but you do need a shared working understanding of what AI is, where it can fail, and what kinds of tasks are appropriate for different AI approaches. This module helps you distinguish pattern recognition, predictive analytics, generative AI, retrieval-augmented generation, and agentic workflows so planning discussions do not treat every AI product as the same kind of intervention.

What You Should Be Able to Do

- Explain AI in plain language to program colleagues, leaders, partners, and community members.
- Identify whether a proposed use is content generation, search and retrieval, prediction, classification, workflow automation, or decision support.
- Name the human role responsible for reviewing, approving, correcting, or rejecting AI-supported outputs.
- Recognize that data quality, workflow design, governance, and public trust are as important as model performance.

Public Health Application

- Use AI concepts to map where an agency already has informal AI use, such as staff drafting messages with public tools or vendors embedding AI into existing platforms.
- Use a common vocabulary during governance review so privacy, IT, program, legal, equity, and communications staff can evaluate the same workflow together.
- Translate technical capabilities into public health value: faster review, better consistency, broader language access, clearer documentation, or earlier signal detection.

Implementation Check

- Before any AI project begins, confirm the purpose, data involved, intended users, public impact, review requirements, and escalation pathway.
- Document what the AI is allowed to do and what decisions remain with qualified public health staff.
- Avoid selecting tools before the department has established vision, readiness, governance, and data-use expectations.

Why This Matters for Your Practice

A shared AI vocabulary prevents agencies from approving tools before they understand the workflow, data, risk level, and accountability structure. This is especially important for STLT public health departments because AI may enter through many doors: staff productivity tools, vendor platforms, analytics systems, communications workflows, or grant-funded pilots.

Reflection Questions

- What AI tools or AI-enabled vendor products are already being used informally?
- Which public health decisions should never be delegated to an AI system?
- Who needs enough AI literacy to participate in governance, procurement, or implementation decisions?

Practical Exercise

Ask each program area to list one current task that involves searching, summarizing, drafting, routing, predicting, translating, or prioritizing. Classify each task by AI type and identify the human decision owner.

Expected Takeaways

- Agency AI vocabulary list
- Initial inventory of known or suspected AI use

- Draft list of decisions that require human accountability

What AI Means for Public Health

Artificial intelligence is a family of methods that enable computer systems to learn from data, recognize patterns, generate new content, and support action toward defined goals. For public health agencies, the most consequential near-term branches are generative AI and agentic AI.

Why This Comes First

Understanding what these systems can and cannot do is the starting point for responsible adoption. Agencies need a shared vocabulary before they can set guardrails, evaluate vendors, select use cases, or explain AI-supported workflows to staff and communities.

Core concepts

- AI is not one technology or one product category.
- Generative AI produces new content from prompts and source material.
- Agentic AI can coordinate a sequence of actions across systems under defined rules.
- Public health use requires human review, privacy protections, equity monitoring, and governance.

References and Resources

- NIST AI Risk Management Framework. Foundational vocabulary for trustworthy AI risk management. https://www.nist.gov/itl/ai-risk-management-framework
- ASPPH AI Framework Report. Public health framing for responsible AI education, research, practice, governance, and workforce preparation. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- WHO: Ethics and governance of artificial intelligence for health. Health-sector principles for ethical and responsible AI use. https://www.who.int/publications/i/item/9789240029200

Generative AI

Generative AI produces new text, images, code, summaries, or synthetic data from prompts and examples.

Generative AI is especially relevant to public health because so much agency work involves drafting, explaining, summarizing, and adapting information for different audiences. It can help create first drafts of education materials, staff communications, grant language, policy summaries, meeting notes, and leadership briefings.

The same fluency that makes generative AI useful also makes it risky. A generated answer can sound polished while including outdated guidance, fabricated details, missing caveats, or language that does not fit the community context. Departments should treat outputs as drafts that require source checking and subject matter review.

For training purposes, staff should practice distinguishing low-risk drafting support from high-risk public-facing or decision-support uses. A safe implementation approach defines approved tools, prohibited data, source requirements, review steps, and documentation expectations before staff use generative AI in routine work.

Module Overview

Generative AI is most useful when public health work requires drafting, summarizing, translating, organizing, or explaining information. It can reduce the time required to create first drafts, compare documents, prepare meeting materials, or convert technical findings into plain language. It should not be used as an unchecked authority on scientific evidence, law, policy, clinical guidance, or public communication.

High-Value Uses

- Draft public health education materials that staff revise for accuracy, tone, literacy level, and cultural context.
- Summarize long guidance documents, inspection notes, surveillance updates, grant requirements, or policy changes for internal review.
- Create first drafts of FAQs, talking points, training materials, meeting agendas, standard operating procedures, and implementation checklists.
- Support grant writing by organizing goals, activities, staffing needs, evaluation measures, and sustainability plans.

Quality Controls

- Require staff review for factual accuracy, citations, public health recommendations, legal requirements, and equity implications.
- Use approved source materials or retrieval-augmented generation when the output must reflect specific agency policy or current guidance.
- Keep protected health information, confidential investigations, sensitive partner data, and restricted records out of public AI tools.
- Label AI-assisted drafts internally so reviewers understand where human verification is needed.

When Not to Use It

- Do not use generative AI to make final eligibility, enforcement, clinical, outbreak response, or public warning decisions.
- Do not rely on unsourced outputs for legal interpretation, epidemiologic conclusions, or scientific claims.
- Do not deploy public-facing chatbots without governance approval, testing, monitoring, and an escalation pathway to staff.

Why This Matters for Your Practice

Generative AI can improve staff efficiency, but it can also produce fluent misinformation, incomplete summaries, biased language, or unsupported recommendations. Public health departments need practical rules for when generative AI can draft, when it can summarize, when it must use approved source material, and when it should not be used at all.

Reflection Questions

- Which materials may be drafted with generative AI and which require approved templates or source documents?
- What human review is required before AI-assisted content is shared with the public, partners, or leadership?
- What data are prohibited from public or consumer AI tools?

Practical Exercise

Take a public health education message, a policy memo, and an internal meeting summary. Define what generative AI could help draft, what source material it would need, and what review steps would be required before use.

Expected Takeaways

- Generative AI acceptable-use rules
- Human review checklist for AI-assisted drafts
- Approved-source guidance for RAG or document-grounded generation

Definition

Generative AI refers to systems trained on large bodies of existing data that can produce new content, including text, images, code, summaries, and synthetic datasets, when prompted by a user.

Large Language Models

The most widely deployed generative AI systems are large language models. They learn statistical patterns across very large text collections and can generate fluent responses to natural-language questions or instructions.

Public health capabilities

- Draft health education materials, vaccination messages, press releases, and program protocols.
- Retrieve and synthesize regulatory guidance, policy documents, and scientific literature using retrieval-augmented generation.
- Translate statistical outputs and surveillance findings into plain-language summaries.
- Write, debug, and explain code for data analysis and visualization.
- Generate realistic but de-identified synthetic datasets for model testing and training.

References and Resources

- WHO: Regulatory considerations on artificial intelligence for health. Health-focused regulatory considerations for AI systems. https://iris.who.int/handle/10665/373421
- NIST Generative AI Profile. Risk management profile for generative AI systems. https://www.nist.gov/itl/ai-risk-management-framework

- ASPPH AI Framework Report. Guidance on human-centered and responsible generative AI use in public health contexts. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- OpenAI Prompt Engineering Guide. Practical guidance on writing clearer prompts, decomposing tasks, using examples, and improving model outputs through structured instructions. https://platform.openai.com/docs/guides/prompt-engineering
- Google Cloud Prompt Engineering Guide. Plain-language overview of prompt engineering concepts, prompt structure, examples, and responsible use considerations. https://cloud.google.com/discover/what-is-prompt-engineering
- Microsoft Prompt Engineering Techniques. Prompting techniques for Azure OpenAI and Microsoft Foundry models, including instructions, grounding, examples, and output formatting. https://learn.microsoft.com/en-us/azure/ai-services/openai/concepts/prompt-engineering

Agentic AI

Agentic AI can coordinate multi-step workflows across systems under explicit rules and human oversight.

Agentic AI introduces a different level of operational complexity because it can coordinate steps rather than simply produce content. A bounded agentic workflow might monitor incoming information, classify it, retrieve relevant documents, create a task, and route it to a staff member. That can be valuable in time-sensitive public health operations, but it also creates new accountability questions.

Departments should train staff to think of agentic AI as workflow infrastructure. The most important design decisions involve boundaries: what the system can access, what actions it can take, when it must stop, when a human must approve, and how every action is logged.

Agentic systems are not appropriate for unsupervised public health decisions. They are best introduced in controlled internal workflows where humans remain responsible for validation, communication, prioritization, and final action.

Module Overview

Agentic AI moves beyond drafting and summarizing by coordinating multiple steps toward a defined goal. In public health, this might mean monitoring incoming reports, gathering relevant data, preparing a draft brief, routing it to the right team, and updating a task queue. Because agentic systems can initiate actions, they require tighter boundaries, audit logs, role-based access, and human approval before consequential steps occur.

Appropriate Starting Points

- Internal workflows where the AI can gather, organize, and route information without contacting the public or making final decisions.
- Surveillance support tasks such as clustering incoming signals, preparing evidence packets, or flagging items for epidemiologist review.
- Administrative workflows such as creating task lists from meeting notes, tracking approvals, or reminding owners about deadlines.
- Care coordination support where staff review all outreach recommendations before any member, client, or patient contact.

Required Controls

- Define allowable actions, prohibited actions, required human approvals, and emergency shutoff procedures.
- Maintain audit logs showing what the AI accessed, generated, changed, routed, or recommended.
- Limit access to the minimum systems and data needed for the workflow.
- Test failure scenarios, including wrong routing, stale data, biased prioritization, hallucinated summaries, and inappropriate escalation.

Public Health Accountability

- A named program or operational owner should remain accountable for the workflow.
- The AI should never obscure who made a public health decision or why a particular action was taken.
- Governance should approve agentic workflows before deployment and review performance after launch.

Why This Matters for Your Practice

Agentic AI can coordinate work across systems, but that makes it higher risk than simple drafting. In public health operations, even routing a task, prioritizing a signal, or preparing an outreach list can affect response time, workload, trust, and equity. Agentic workflows need explicit boundaries before they are piloted.

Reflection Questions

- What actions may the AI take without approval, and what actions require human approval?
- What audit logs are needed to reconstruct what happened?
- How will staff pause, override, or escalate an agentic workflow?

Practical Exercise

Map a multi-step workflow such as outbreak signal review, inspection report preparation, or care coordination outreach. Mark which steps the AI may support, which steps require human approval, and which steps are prohibited.

Expected Takeaways

- Agentic workflow boundary statement
- Human approval and escalation map
- Audit log and monitoring requirements

Definition

Agentic AI systems can analyze information, generate outputs, and take a sequence of actions toward a defined goal across multiple systems while following explicit rules and retaining human oversight.

Why It Matters

Agentic capabilities are most useful where public health work already involves multi-step workflows: gathering evidence, monitoring information streams, routing tasks, drafting updates, and handing decisions back to qualified staff.

Practical agentic uses

- Pull data from multiple systems, summarize findings, and route a draft brief for epidemiologist review.
- Monitor incoming news feeds or case reports, classify and prioritize items, and create surveillance task queues.
- Draft follow-up messages to patients or partners and queue them for staff approval.
- Coordinate dashboards, case management systems, and communication platforms when predefined thresholds are crossed.

References and Resources

- NIST AI Risk Management Framework. Useful for defining boundaries, oversight, monitoring, and accountability for agentic systems. https://www.nist.gov/itl/ai-risk-management-framework
- WHO: Ethics and governance of artificial intelligence for health. Health-sector governance principles for systems that can influence decisions or workflows. https://www.who.int/publications/i/item/9789240029200
- ASPPH AI Framework Report. Public health governance, human oversight, and AI literacy context for agentic workflows. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/

Deep Research

Deep Research tools autonomously plan multi-step research tasks, gather sources, and draft citation-based reports for expert review.

Deep Research tools can help departments organize complex evidence, policy, or funding questions into structured summaries. They are useful when staff need a first-pass scan of literature, agency guidance, jurisdictional policies, implementation examples, or grant requirements.

These tools should not be treated as substitutes for professional judgment or formal evidence review. Staff still need to verify citations, check dates, assess source quality, and decide whether findings apply to the department's population, authority, and operational context.

A strong training approach asks staff to evaluate a generated research brief: Are the sources real and current? Are claims supported? Are limitations stated? Does the summary separate evidence from opinion or vendor claims? What decisions could this support, and what review is still required?

Module Overview

Deep Research tools can help public health departments move faster when they need a structured scan of literature, policies, funding requirements, legal materials, program models, or emerging issues. These tools are best treated as research assistants that produce a starting synthesis. Staff still need to verify sources, check dates, resolve conflicting evidence, and decide what applies to the department's context.

Useful Assignments

- Scan peer-reviewed and agency sources on an emerging disease, intervention, risk factor, or public health technology.
- Compare AI policies, procurement requirements, or governance models across jurisdictions.
- Summarize grant notices, eligibility requirements, evaluation expectations, and allowable costs.
- Prepare a briefing that separates strong evidence, emerging evidence, implementation examples, and unanswered questions.

Review Standards

- Require source links and citations for every consequential claim.
- Check publication dates, jurisdiction, population relevance, and whether the source is primary, secondary, vendor-authored, or opinion-based.
- Have subject matter experts review conclusions before they inform policy, funding, public communication, or operational decisions.
- Document what sources were included or excluded when the synthesis supports governance or leadership decisions.

Actionable Output

- Ask for a decision-ready format: summary, implications for STLT public health, risks, resource needs, equity considerations, and recommended next steps.
- Convert the research output into a tool artifact such as a policy scan, gap register, governance memo, funding brief, or implementation roadmap.
- Update scans when guidance, funding rules, laws, or evidence change.

Why This Matters for Your Practice

Deep Research tools can make literature scans and policy reviews faster, but public health departments still need defensible evidence standards. A generated synthesis is only useful if sources are current, relevant, independently verifiable, and interpreted by staff who understand the public health context.

Reflection Questions

- What source types are acceptable for the decision being supported?
- Who verifies citations, dates, jurisdictional relevance, and evidence strength?
- How will the department distinguish evidence, implementation examples, vendor claims, and unresolved questions?

Practical Exercise

Use a current public health topic and design a research brief template with sections for evidence summary, source quality, STLT relevance, equity considerations, operational implications, and open questions.

Expected Takeaways

- AI-assisted research brief template
- Source verification checklist
- Policy or literature scan protocol

A Practical Agentic Capability

Deep Research tools autonomously plan and execute multi-step research tasks, gather and cross-reference sources, and generate citation-based reports with transparent reasoning for expert review.

Best Fit

These tools are well suited for first drafts of literature synthesis, policy and legal scans, strategic planning, scenario analysis, and comparative program reviews. They should accelerate expert work, not replace expert validation.

Appropriate tasks

- Literature synthesis on emerging threats or intervention strategies.
- Policy and legal scans across jurisdictions.
- Strategic planning and scenario analysis.
- Comparative analysis of public health programs, interventions, or regulations.

References and Resources

- NIH PubMed. Primary source for biomedical and public health literature searches. https://pubmed.ncbi.nlm.nih.gov/
- CDC Publications and Resources. Public health reports, guidance, and evidence sources for staff review. https://www.cdc.gov/publications/
- ASPPH AI Framework Report. Framework language for responsible use of AI-supported research and synthesis. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/

AI-Supported Public Health Workflows

AI can support signal detection, evidence gathering, analysis, synthesis, briefing, communication, and decision support.

Public health AI should start with workflows, not tools. A workflow lens helps teams identify where work is slow, repetitive, inconsistent, or difficult to coordinate, and where AI might help without replacing public health judgment.

Examples include summarizing incoming reports, detecting unusual patterns, drafting situation updates, routing tasks, preparing evidence packets, or tracking implementation deadlines. Each potential use should be evaluated in context: who uses the output, what decision follows, what could go wrong, and what evidence would show improvement.

Before redesigning a workflow around AI, teams should complete basic business process analysis: define the current-state process, identify actors and handoffs, document information flows, locate bottlenecks and rework, and clarify what the future-state process should accomplish. This prevents teams from automating a broken workflow or adding AI before the operational problem is well understood.

Your training should help staff map a real workflow from trigger to output. Once the workflow is visible, the team can decide which steps AI may support, which require human review, and which should remain outside AI support altogether.

Module Overview

AI is most useful when it is tied to a specific public health workflow rather than introduced as a general productivity tool. Workflow mapping helps teams identify where AI could support signal detection, evidence gathering, analysis, synthesis, communication, decision support, documentation, monitoring, or service delivery without displacing accountability.

Workflow Mapping Questions

- What event starts the workflow, and what output ends it?
- Where do staff spend time searching, summarizing, routing, checking, rewriting, or reconciling information?
- Which steps require judgment, legal authority, community relationship, scientific interpretation, or public accountability?
- What data systems, forms, emails, dashboards, or documents are involved?

Potential AI Support

- Detect unusual patterns and prepare alerts for human review.
- Summarize reports, partner updates, inspection notes, or case records into structured fields.
- Draft situation reports, FAQs, or internal updates from approved source material.
- Track implementation tasks, deadlines, approvals, and follow-up actions across teams.

Design Safeguards

- Keep AI support close to the staff workflow and test whether it actually saves time or improves quality.
- Measure false positives, false negatives, staff burden, equity effects, and downstream operational impact.
- Make handoffs explicit so staff know when an AI output is a draft, a flag, a recommendation, or a required review item.

Why This Matters for Your Practice

AI projects often fail when they are not anchored in a real workflow. Public health work includes handoffs, legal authorities, partner relationships, community trust, and time-sensitive decisions. Workflow mapping helps agencies find where AI can reduce burden while preserving judgment and accountability.

Reflection Questions

- What problem in the workflow is AI expected to solve?
- Where does the workflow require professional judgment, legal authority, or community knowledge?
- How will staff know whether AI improves timeliness, quality, consistency, equity, or workload?

Practical Exercise

Choose one workflow and map the trigger, inputs, staff roles, decisions, outputs, handoffs, risks, and performance measures. Identify one AI support point and one step that should remain fully human-controlled.

Expected Takeaways

- AI workflow map
- Workflow risk and control table
- Pilot success measures

Common Public Health Workflow

Many public health activities move from signal detection to evidence gathering, analysis, synthesis, briefing, communication, and decision support. Mapping AI to workflow stages helps agencies choose bounded, reviewable uses.

Reference Table

- Signal Detection - Identify signals from surveillance systems, laboratory reporting, emergency departments, or environmental monitoring. - Anomaly detection and LLM-assisted monitoring of open-source information streams.
- Evidence Gathering - Collect evidence to determine whether a signal represents a meaningful concern. - Agentic systems retrieve datasets, compile literature, and assemble supporting documentation.
- Analysis - Evaluate collected information to determine significance. - AI-supported statistical analysis, exploratory visualization, and pattern identification.
- Synthesis - Combine information from multiple sources into a coherent picture. - Generative AI summarizes findings, consolidates reports, and drafts evidence summaries.
- Briefing and Communication - Communicate findings through dashboards, situation reports, policy briefs, or advisories. - Generative AI drafts reports, summarizes trends, and adapts messages for audiences.
- Decision Support - Use evidence to guide response activities and policy decisions. - AI organizes relevant information and produces structured briefing materials for leadership review.

References and Resources

- CDC Data Modernization Initiative. Context for modern public health data flows and AI-ready infrastructure. https://www.cdc.gov/surveillance/data-modernization/index.html
- CDC STLT Data Connection. Peer-learning resource for public health data modernization and workflow improvement. https://www.cdc.gov/data-modernization/php/stlt-data-connection/index.html
- NIST AI Risk Management Framework. Framework for mapping workflow risks, controls, and monitoring. https://www.nist.gov/itl/ai-risk-management-framework

Risks, Limitations, and Guardrails

Public health AI risks include hallucination, bias, privacy exposure, automation bias, model drift, and performance loss on complex tasks.

AI risk in public health is broader than technical model performance. A system can be accurate in a test setting but still create harm if it exposes sensitive data, increases inequity, produces unclear recommendations, encourages over-trust, or undermines community confidence.

Risk management should be practical and repeatable. You should learn to ask what harm could occur, who could be affected, how likely the harm is, how it would be detected, and who has authority to pause or correct the system.

This module should prepare staff to participate in risk review discussions. The goal is not to eliminate all risk, but to make risk visible, assign ownership, document mitigations, and monitor whether safeguards work after deployment.

Module Overview

AI risks in public health are not limited to technical errors. They include privacy exposure, hallucinated guidance, biased prioritization, automation bias, model drift, vendor opacity, accessibility barriers, environmental burden, and loss of public trust. Risk management should begin before tool selection and continue through deployment and monitoring.

Common Risk Categories

- Accuracy risk: the output is wrong, incomplete, out of date, or unsupported by evidence.
- Equity risk: the system performs worse for a subgroup, language community, rural area, disability group, Tribal community, or underrepresented population.
- Privacy and security risk: sensitive data are exposed to an inappropriate tool, vendor, model, or user.
- Operational risk: staff over-trust the system, workarounds emerge, or responsibilities become unclear.
- Transparency risk: the public cannot understand when AI is used, what humans decide, or how to raise concerns.

Mitigation Practices

- Use risk tiers so higher-impact systems receive deeper review, documentation, validation, and monitoring.
- Require human review, source verification, and documented decision ownership.
- Test outputs with realistic public health examples before use.
- Monitor errors, incidents, complaints, subgroup performance, and drift after deployment.

Governance Triggers

- Escalate to governance when AI affects public communication, eligibility, enforcement, surveillance prioritization, resource allocation, clinical or care coordination decisions, or community trust.
- Pause or retire systems when risk controls fail, performance changes, or the use no longer aligns with agency principles.

Why This Matters for Your Practice

Responsible AI risk management is not a one-time checklist. Public health risks change as data, models, vendors, disease patterns, staff behavior, and community needs change. Agencies need repeatable risk review methods that can be used before approval and after deployment.

Reflection Questions

- What harms could occur if the AI output is wrong, biased, delayed, misunderstood, or over-trusted?
- What data, populations, or workflows are most vulnerable to error?
- What conditions should trigger pause, review, correction, or retirement?

Practical Exercise

Select a proposed AI use case and identify risks across accuracy, privacy, security, equity, transparency, workforce burden, legal authority, and public trust. Define one mitigation for each risk.

Expected Takeaways

- Risk register
- Mitigation plan
- Incident escalation criteria

Risk Is Operational

Generative and agentic AI risks are especially consequential in public health because errors can affect trust, equity, privacy, resource allocation, and public communication.

Limitations, Risks, and Mitigations

- Hallucination - AI can generate plausible-sounding but incorrect information. - Require human adjudication before consequential action.
- Performance degradation with complexity - Accuracy can drop as tasks become more complex or ambiguous. - Match AI tools to bounded tasks and validate before deployment.
- Data and representational bias - Models trained on non-representative data can underperform for underrepresented populations. - Assess data provenance and monitor subgroup performance continuously.
- Privacy and data sovereignty - Sending protected or sensitive data to unsuitable tools can violate HIPAA or tribal data sovereignty principles. - Use appropriate legal agreements, access controls, and approved environments.
- Automation bias - Staff may reduce critical scrutiny after working with AI outputs over time. - Use audits, review checkpoints, training, and governance reporting.

References and Resources

- NIST AI Risk Management Framework. Core framework for identifying, measuring, managing, and governing AI risk. https://www.nist.gov/itl/ai-risk-management-framework
- HHS Civil Rights and AI. Civil rights and nondiscrimination considerations for AI in health and human services. https://www.hhs.gov/civil-rights/for-individuals/special-topics/ai/index.html
- WHO: Ethics and governance of artificial intelligence for health. Health AI risk, equity, transparency, and accountability guidance. https://www.who.int/publications/i/item/9789240029200

AI as a Public Health Systems Shift

AI changes public health infrastructure, workforce expectations, policy, data systems, procurement, public trust, communications, surveillance, and service delivery.

AI adoption changes the operating system of a public health department. It affects data infrastructure, workforce expectations, procurement, governance, legal review, public communication, quality improvement, emergency response, and community trust.

When AI is treated as a one-off tool purchase, departments can end up with disconnected pilots, uneven safeguards, unclear ownership, and unsustainable costs. A systems-shift approach connects AI work to existing modernization, equity, workforce, and accountability priorities.

Your training should help leaders and staff see dependencies. A use case may require governance approval, data quality improvements, staff training, vendor review, funding, public notice, and monitoring before it can responsibly move forward.

Module Overview

AI adoption is a systems change for public health departments. It affects data modernization, procurement, workforce roles, governance authority, legal review, community engagement, communications, emergency response, program operations, and evaluation. Treating AI as a one-time technology purchase leaves agencies with fragmented pilots and unmanaged risk.

Systems Affected

- Data systems need quality checks, access rules, documentation, interoperability, and appropriate AI environments.
- Workforce systems need training, role clarity, bridge professionals, change support, and leadership expectations.
- Governance systems need intake, risk review, approval, monitoring, incident response, and policy updates.
- Community systems need transparency, plain-language explanation, feedback pathways, and attention to missing voices.

Planning Implications

- Sequence adoption through plays rather than approving isolated pilots.
- Connect AI planning to data modernization, workforce development, funding, procurement, and quality improvement.
- Identify where existing policies can be extended and where AI-specific standards are needed.
- Avoid pilots that cannot be sustained, monitored, evaluated, or governed.

Leadership Role

- Leaders should set the expectation that responsible AI is mission infrastructure, not a side project.
- Executive sponsors should ensure governance has authority, staff have time, and public accountability is built into decisions.

Why This Matters for Your Practice

AI changes public health infrastructure, not just individual tasks. Departments need to align AI adoption with data modernization, workforce development, procurement, legal review, community engagement, funding, and evaluation so AI does not become a series of disconnected pilots.

Reflection Questions

- Which agency systems must change for AI to be governed and sustained?
- Where do existing policies already apply, and where are AI-specific standards needed?
- How will leadership coordinate AI work across programs rather than approving isolated tools?

Practical Exercise

Create a systems map showing how AI adoption affects governance, data, technology, workforce, procurement, communications, equity, and funding. Identify gaps that must be addressed before pilots scale.

Expected Takeaways

- AI systems-change map
- Cross-program implementation dependencies
- Leadership decision agenda

Beyond Tool Adoption

AI should be treated as a public health systems shift. It affects how agencies govern information, structure work, communicate with the public, procure technology, train staff, and maintain trust.

Why This Matters for STLTs

State, tribal, local, and territorial public health departments often operate under constrained budgets, changing legal requirements, distributed authority, and high public accountability. Fragmented AI adoption can create uneven privacy protections, unclear ownership, duplicated vendor purchases, and inconsistent public explanations.

Systems affected by AI

- Governance and decision authority.
- Data modernization, interoperability, and data quality.
- Workforce roles, AI literacy, and change management.
- Procurement, vendor oversight, cybersecurity, and public records obligations.
- Community trust, transparency, and public accountability.
- Monitoring, incident reporting, continuous improvement, and system retirement.

References and Resources

- ASPPH AI Framework Report. Frames AI as a public health systems issue rather than a narrow technology purchase. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- CDC STLT resources. State, tribal, local, and territorial public health implementation context. https://www.cdc.gov/public-health-gateway/php/index.html
- Public Health Infrastructure Grant. Workforce, data systems, organizational capacity, and equity investments that can support responsible AI readiness. https://www.cdc.gov/infrastructure-phig/php/index.html

AI as a Digital Determinant of Health

AI can shape access to services, quality of information, exposure to misinformation, eligibility decisions, language access, public benefits, and community trust.

AI can influence health by shaping digital access to information, services, outreach, and decision pathways. If an AI-supported process is hard to understand, unavailable in needed languages, inaccessible to people with disabilities, or based on incomplete data, it can widen existing inequities.

Departments should evaluate AI as part of the broader digital environment people must navigate. A chatbot, risk score, translation tool, scheduling system, or eligibility-support process may affect whether someone receives timely, understandable, and fair support.

Your training should ask staff to identify who may be excluded or burdened by a proposed AI use. The answer should inform design, engagement, alternatives, and monitoring before launch.

Module Overview

AI can become a digital determinant of health because it shapes how people receive information, access services, are prioritized for outreach, experience language access, or are affected by automated decisions. Departments should examine whether AI-supported systems make services more reachable and fair, or whether they create new barriers.

Where Digital Determinants Appear

- Chatbots, eligibility tools, translation systems, online appointment systems, risk scores, outreach lists, public dashboards, and complaint intake tools.
- Public health messaging that is generated, translated, personalized, or targeted with AI.
- Vendor systems that embed AI into case management, inspection, contact center, or analytics platforms.

Equity Review Questions

- Who benefits from the tool, and who might be excluded because of language, disability, broadband, device access, trust, literacy, geography, or data representation?
- Does the tool work for rural, Tribal, immigrant, older adult, disability, low-literacy, and limited-English-proficiency communities?
- Can people receive help, appeal errors, correct information, or use a non-AI pathway?

Action Steps

- Include digital access and language access in use case prioritization.
- Test public-facing outputs with affected users before launch.
- Monitor service access, complaints, correction requests, and subgroup outcomes after deployment.

Why This Matters for Your Practice

AI can influence whether people receive understandable information, timely services, fair prioritization, language access, or meaningful alternatives. Departments should treat AI-supported workflows as part of the digital environment that shapes health access and trust.

Reflection Questions

- Who might be excluded by the data, interface, language, channel, or workflow?
- Does the AI-supported process create or reduce barriers to services?
- What non-digital or non-AI alternative is available for important services?

Practical Exercise

Review a public-facing AI use case and identify access barriers related to language, disability, broadband, device access, literacy, rural context, trust, and data representation.

Expected Takeaways

- Digital access impact review
- Language and accessibility mitigation plan
- Alternate pathway documentation

AI and Digital Determinants of Health

AI can become part of the digital environment that affects health and access to public health services. It can influence who receives information, how quickly services are delivered, which languages are supported, which risks are surfaced, and which communities are overlooked.

Public Health Implications

A tool that improves efficiency for one group may create new barriers for another. Agencies should review AI-enabled workflows for broadband access, digital literacy, disability access, language access, immigration-related trust concerns, rural context, tribal data sovereignty, and misinformation exposure.

Questions to ask

- Who benefits from this AI-supported workflow and who may be excluded?
- Does the workflow depend on digital access, English proficiency, stable housing, or complete records?
- Could AI outputs affect access to services, benefits, outreach, enforcement, or public guidance?
- How will the agency detect inequitable effects after launch?

References and Resources

- HHS Civil Rights and AI. Civil rights context for AI-enabled access, language, disability, and service equity concerns. https://www.hhs.gov/civil-rights/for-individuals/special-topics/ai/index.html
- ASPPH AI Framework Report. Human-centered AI, missing voices, public trust, and equity framing. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- CDC Health Equity. Public health equity context for assessing AI-supported workflows. https://www.cdc.gov/healthequity/

Human-Centered AI

Human-centered AI keeps you and other public health professionals, affected communities, and accountable decision-makers at the center of design, review, and action.

Human-centered AI keeps people at the center of design and accountability. In public health, this includes staff who use the system, leaders who are responsible for decisions, partners who depend on agency information, and communities affected by the workflow.

The purpose of human-centered AI is not simply to keep a human somewhere in the loop. It is to define meaningful human authority: who reviews outputs, who can override recommendations, who explains decisions, and how people can raise concerns or correct errors.

Your training should focus on decision rules. For every AI-supported workflow, you should be able to say what AI may do, what it may not do, what humans must approve, and how concerns will be escalated.

Module Overview

Human-centered AI begins with the people affected by a workflow: public health staff, community members, partners, and decision-makers. The goal is not to automate as much as possible. The goal is to improve public health work while preserving human judgment, relationships, dignity, explainability, and accountability.

Design Principles

- Start with a public health problem and affected users, not with a tool.
- Define what humans must decide, approve, explain, or correct.
- Make AI outputs reviewable, editable, and contestable.
- Design for the staff who will use the system and the communities who may be affected by it.

Workflow Practices

- Name the decision owner for every AI-supported workflow.
- Require expert review before public-facing communication, enforcement, resource allocation, or clinical/care coordination action.
- Document correction, appeal, complaint, and escalation pathways.
- Evaluate whether the tool reduces burden or simply moves burden to different staff or communities.

Community Accountability

- Use engagement to identify concerns before launch, especially for high-impact or community-facing systems.
- Explain AI use in plain language and invite feedback.
- Avoid deploying systems where affected people cannot understand or challenge the outcome.

Why This Matters for Your Practice

Human-centered AI keeps public health purpose, staff expertise, and community accountability at the center. This is especially important when AI affects outreach, communication, prioritization, inspections, surveillance, or services.

Reflection Questions

- Who is affected by the workflow and who has been consulted?
- What decision remains with a named human owner?
- How can people ask questions, correct information, or challenge an outcome?

Practical Exercise

For one proposed use case, write a human decision rule that defines what AI may support, what it may recommend, what a human must approve, and what the AI may not do.

Expected Takeaways

- Human decision rule
- User and community needs summary
- Appeal, correction, or feedback pathway

Definition

Human-centered AI is designed to assist people, not displace accountability. Public health staff, communities, and accountable leaders remain responsible for interpretation, judgment, relationship-building, and consequential decisions.

Operational Rule

For each use case, define what AI may do, what it may not do, who reviews outputs, who makes the final decision, how decisions are documented, and how people can challenge or correct an AI-supported outcome.

Required safeguards

- Named human owner for every consequential decision.
- Human review before public communication, enforcement, service denial, escalation, or clinical/public health action.
- Audit trail showing AI output, human review, decision, and rationale.
- Correction or appeal pathway when people are affected.
- Monitoring for automation bias and over-trust.

References and Resources

- Blueprint for an AI Bill of Rights. Principles for automated systems that preserve human agency, notice, explanation, and safeguards. https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/
- NIST AI Risk Management Framework. Useful for defining human oversight, decision accountability, and risk controls. https://www.nist.gov/itl/ai-risk-management-framework
- ASPPH AI Framework Report. Public health-specific framing for human-centered AI and accountable implementation. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/

AI Literacy for Public Health Staff

AI literacy means you can recognize AI use, understand major failure modes, protect sensitive data, communicate uncertainty, document use, and know when not to use AI.

AI literacy is a baseline workforce skill for responsible adoption. You may not need the same level of technical expertise, but everyone using or overseeing AI should understand approved uses, prohibited data, common failure modes, and escalation expectations.

Role-based literacy matters. Executives need to understand governance and risk. Program colleagues need workflow and review guidance. Analysts need validation and monitoring practices. Procurement staff need vendor and contract questions. Communications staff need standards for public-facing content.

Your training should give staff practical confidence: how to use approved tools, how to protect sensitive data, how to check outputs, how to document AI assistance, and when to stop and ask for review.

Module Overview

AI literacy is a workforce capability. You should know what AI is, what it is approved for, what data may be used, how outputs can fail, how to document AI use, and when to escalate concerns. Literacy expectations should be role-based because leaders, program colleagues, analysts, IT, privacy, procurement, communications, and governance members have different responsibilities.

Core Competencies

- Recognize approved and unapproved AI use.
- Protect PHI, confidential, restricted, and sensitive data.
- Identify hallucination, bias, automation bias, data quality problems, and inappropriate use.
- Use prompts, source materials, and review steps responsibly.
- Document AI-assisted work when required by agency policy.

Role-Based Training

- Executives need governance, risk, resource, and public accountability orientation.
- Program colleagues need workflow, privacy, prompt, and review guidance.
- Analysts need validation, documentation, model monitoring, and data quality practices.
- Communications staff need plain-language, misinformation, translation, and public review standards.
- Procurement and legal staff need vendor, contract, audit, and data-rights requirements.

Sustainment

- Refresh training as tools, policies, and risks change.
- Create reporting pathways for staff concerns and near misses.
- Use AI champions or bridge professionals to support day-to-day adoption.

Why This Matters for Your Practice

AI literacy is necessary for safe use at every level of the department. Staff need role-appropriate knowledge before they are asked to use AI tools, review outputs, approve vendors, participate in governance, or explain AI to the public.

Reflection Questions

- What does each role need to know to use or oversee AI responsibly?
- How will the department train new staff and refresh training over time?
- How will staff report concerns, near misses, or unsafe outputs?

Practical Exercise

Create a role-based training matrix for executives, program colleagues, analysts, IT, privacy, procurement, communications, and governance members.

Expected Takeaways

- Role-based AI literacy matrix
- Training plan
- Staff concern reporting pathway

What AI Literacy Means

AI literacy is the ability to recognize AI use, understand what a system can and cannot do, evaluate outputs critically, protect sensitive information, communicate uncertainty, and know when AI should not be used.

Role-Based Expectations

All staff need basic awareness, but supervisors, communications staff, epidemiologists, data analysts, procurement staff, IT/security staff, legal/privacy staff, equity staff, and governance members need different depth and practice examples.

Core staff competencies

- Recognize generative, predictive, and agentic AI use.
- Avoid entering PHI, confidential, or restricted data into unapproved tools.
- Check outputs for hallucinations, bias, missing context, and unsupported claims.
- Document AI assistance when required by policy.
- Escalate concerns through governance or incident reporting pathways.

References and Resources

- ASPPH AI Framework Report. AI literacy, bridge professional, and workforce preparation concepts for public health. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- Public Health Foundation TRAIN Learning Network. Training platform for public health workforce development. https://www.train.org/
- CDC Public Health Workforce Development. Public health workforce training and capability-building context. https://www.cdc.gov/workforce/

Automation Bias and Over-Trust

Automation bias occurs when people accept AI outputs too readily because they appear confident, precise, or authoritative.

Automation bias occurs when people accept AI outputs too readily because they appear confident, precise, or efficient. In public health, this can affect surveillance triage, inspection documentation, outreach prioritization, public messaging, or leadership briefings.

Your training should make critical review a normal part of AI-supported work. You should learn to look for missing context, unsupported claims, outdated assumptions, subgroup effects, and uncertainty that may not be obvious in a polished output.

Departments can reduce automation bias by requiring review prompts, second checks for high-impact outputs, decision-owner sign-off, override documentation, and periodic audits of AI-assisted work.

Module Overview

Automation bias occurs when people give too much weight to AI output because it appears confident, precise, or authoritative. In public health, this can affect surveillance review, risk scoring, inspection documentation, prioritization, public messaging, or resource decisions. Deliberate review practices are needed so AI supports, rather than substitutes for, professional judgment.

Warning Signs

- Staff stop checking sources because AI summaries are usually fluent.
- Risk scores become the primary basis for decisions without review of context.
- Draft messages are approved without checking public health guidance, equity implications, or local context.
- A workflow lacks a named human decision owner.

Controls

- Use second-review requirements for high-impact outputs.
- Require confidence, limitation, source, and uncertainty notes in briefs.
- Separate AI recommendations from human decisions in documentation.
- Train staff to look for omissions, unsupported claims, and plausible but wrong details.

Monitoring

- Audit samples of AI-assisted work.
- Track overrides, corrections, incidents, complaints, and false positives or false negatives.
- Review whether staff are using the tool as intended or developing unsafe shortcuts.

Why This Matters for Your Practice

Automation bias is a practical operational risk. Staff may over-trust outputs that are fluent, precise, or repeated often. Public health workflows need review steps that make skepticism normal and expected.

Reflection Questions

- Where might staff accept AI output too quickly?
- What second-review or sign-off is needed for high-impact outputs?
- How will overrides, corrections, and disagreements be documented?

Practical Exercise

Take one AI-assisted output and design a review form that asks staff to check sources, uncertainty, missing context, subgroup impact, and whether the recommendation should be accepted, modified, or rejected.

Expected Takeaways

- Automation bias review checklist
- Decision owner sign-off process
- Override and correction log

Definition

Automation bias occurs when people over-rely on AI outputs because the system appears fast, confident, technical, or authoritative. Over-trust can be especially risky in high-pressure public health settings.

Prevention

Workflows should require reviewers to check source evidence, local context, limitations, uncertainty, and subgroup impacts. Reviewers should have permission and time to disagree with AI outputs.

Design controls

- Display limitations and uncertainty with AI outputs.
- Require evidence links or source material for factual summaries.
- Use second review for high-impact decisions.
- Track cases where staff override or correct AI outputs.
- Train staff to ask what evidence would change the recommendation.

References and Resources

- NIST AI Risk Management Framework. Risk controls for over-trust, human review, testing, and monitoring. https://www.nist.gov/itl/ai-risk-management-framework
- HHS Civil Rights and AI. Equity and civil rights considerations when automated systems influence decisions. https://www.hhs.gov/civil-rights/for-individuals/special-topics/ai/index.html
- WHO: Ethics and governance of artificial intelligence for health. Health AI ethics guidance addressing human autonomy and oversight. https://www.who.int/publications/i/item/9789240029200

Missing Voices in Data

Missing voices analysis asks whose experiences, records, language needs, disability access needs, rural context, tribal sovereignty interests, or community knowledge are absent from the data.

Missing voices analysis asks whose experiences, data, language needs, disability access needs, geography, or community knowledge are absent from the design and evaluation of an AI-supported workflow. It is a practical equity tool, not a theoretical exercise.

A model may perform well overall while failing groups that are underrepresented or misclassified. A communication tool may be readable for some audiences but not others. A prioritization workflow may reflect historical service patterns rather than current need.

Your training should help staff identify missing voices early, decide who should be consulted, and define how the department will monitor whether the AI-supported workflow works across affected communities.

Module Overview

Missing voices analysis asks who is absent from the data, design process, validation testing, governance discussion, or evaluation. Public health AI can look accurate overall while failing groups that are underrepresented, misclassified, or affected differently by the workflow. Missing voices review should happen before use case approval and again during monitoring.

Who May Be Missing

- People with limited English proficiency, people with disabilities, rural residents, people experiencing homelessness, immigrant communities, Tribal communities where applicable, older adults, children, justice-involved populations, and groups with limited digital access.
- Frontline staff who understand workflow realities but are not included in planning.
- Community partners who can identify trust, access, or cultural context concerns.

How to Apply It

- Review whether source data include the populations affected by the decision or communication.
- Ask whether subgroup performance will be tested before and after deployment.
- Use community validation when the workflow has public-facing or community-impacting consequences.
- Document what groups were consulted, what concerns were raised, and how the design changed.

Ongoing Monitoring

- Track subgroup performance, service access, complaints, corrections, and unintended impacts.
- Revisit the analysis when the model, data, policy, population, or workflow changes.

Why This Matters for Your Practice

Missing voices analysis helps agencies avoid designing AI systems around only the data and perspectives that are easiest to obtain. It is essential for equity, trust, and real-world performance.

Reflection Questions

- Whose data, experience, language needs, disability access needs, rural context, or community knowledge may be missing?
- Who should validate assumptions before launch?
- How will subgroup performance and community concerns be monitored after deployment?

Practical Exercise

For a selected use case, identify groups that may be missing from the data, planning process, validation sample, or governance review. Define how each gap will be addressed.

Expected Takeaways

- Missing voices analysis
- Community validation plan
- Subgroup monitoring measures

Definition

Missing voices analysis identifies people, communities, settings, and forms of knowledge that are absent or poorly represented in the data, design process, validation evidence, or monitoring plan.

Why It Matters

An AI system may perform acceptably in aggregate while failing for small populations, rural communities, tribal communities, immigrants, people with disabilities, people with limited English proficiency, people experiencing homelessness, or communities with incomplete records.

Review focus

- Groups missing or underrepresented in source data.
- Language, accessibility, and digital access barriers.

- Tribal sovereignty and community data governance concerns.
- Historical underreporting, misclassification, and mistrust.
- Subgroup validation and monitoring plans.

References and Resources

- ASPPH AI Framework Report. Introduces missing voices and human-centered AI considerations for public health. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- CDC Health Equity. Health equity context for identifying who may be missed or harmed by public health systems. https://www.cdc.gov/healthequity/
- NIH Tribal Health Research Office. Tribal data sovereignty and tribal health research context. https://dpcpsi.nih.gov/thro

Public Transparency and Community Trust

Public health agencies should be able to explain when AI is used, why it is used, what data are involved, what humans decide, and how concerns can be raised.

Transparency is central to public trust. Health departments should be able to explain when AI is used, why it is used, what data are involved, what humans decide, what safeguards exist, and how people can ask questions or raise concerns.

Transparency should be proportionate to the impact of the AI use. An internal drafting tool may need staff-facing documentation, while a public-facing chatbot, prioritization tool, or outreach workflow may require public notice, FAQs, and feedback pathways.

Your training should help staff write clear explanations without overstating AI capability. A useful notice explains the workflow in plain language and names the human accountability structure.

Module Overview

Transparency helps protect public trust. Public health departments should be able to explain when AI is used, why it is used, what data are involved, what humans decide, what safeguards exist, and how people can raise concerns or seek correction. Transparency should be proportional to risk and understandable to the affected audience.

What to Explain

- The public health purpose of the AI-supported activity.
- Whether AI drafts, summarizes, prioritizes, routes, predicts, translates, or recommends.
- What data types are involved and what data are not used.
- Who reviews outputs and who makes final decisions.
- How people can ask questions, report concerns, or request correction.

Formats

- Public AI use notices for community-facing systems.
- FAQs for staff, partners, and affected communities.
- Plain-language summaries for governing boards, advisory committees, and public meetings.
- Internal records in the AI inventory and governance documentation.

Trust-Building Practices

- Be clear about limitations and human oversight rather than overstating AI capability.
- Avoid vague claims such as 'AI-powered' without explaining what the system actually does.
- Update notices when use, data, vendor, or risk level changes.

Why This Matters for Your Practice

Public trust depends on whether agencies can explain AI use clearly. Transparency should not be a vague notice that AI exists; it should explain purpose, data, human review, limits, safeguards, and feedback options.

Reflection Questions

- What should staff, partners, and the public know about this AI use?

- What information can be shared without compromising privacy or security?
- How will notices be updated when the AI use changes?

Practical Exercise

Draft a plain-language AI use notice for one public-facing workflow. Include purpose, what AI does, what humans decide, data types, safeguards, limitations, and contact or correction options.

Expected Takeaways

- Plain-language AI use notice
- Public FAQ
- Feedback and correction process

Plain-Language Transparency

Public health agencies should be able to explain when AI is used, what purpose it serves, what data are involved, what humans still decide, what the system cannot do, and how questions or concerns can be raised.

Trust and Accountability

Transparency is not a marketing statement. It is part of accountability, especially when AI-supported workflows affect public communication, eligibility, prioritization, surveillance, inspections, outreach, or community-facing decisions.

Public notice elements

- What AI-supported system or workflow is being used.
- Why the department is using it.
- What data are used and what data are not used.
- How human review and final decision-making work.
- Known limitations, risks, and safeguards.
- How people can ask questions, request help, or report concerns.

References and Resources

- Blueprint for an AI Bill of Rights. Notice, explanation, and alternatives principles for automated systems. https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/
- ASPPH AI Framework Report. Public transparency, trust, and community accountability expectations. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- NIST AI Risk Management Framework. Governance and transparency concepts for trustworthy AI. https://www.nist.gov/itl/ai-risk-management-framework

Tiered Data Use for AI

Tiered data use classifies information by sensitivity and defines which AI environments are appropriate for public, internal, confidential, and highly restricted data.

Tiered data use turns privacy and security principles into practical rules. Staff need to know which data can be used in public tools, enterprise tools, restricted environments, or only within tightly governed systems.

Public health departments handle many categories of information, including public materials, internal drafts, confidential operational records, protected health information, partner data, and data governed by specific community or Tribal authorities. Each category may require different approvals and safeguards.

Your training should use realistic scenarios so staff can classify data correctly and choose the appropriate AI environment. The goal is to prevent accidental disclosure and to make safe use easier than unsafe use.

Module Overview

Tiered data use gives staff practical rules for which AI environments can be used with which kinds of information. Public, internal, confidential, protected, restricted, and sovereign data may require different tools, approvals, vendors, contracts, security controls, and documentation. Without tiers, staff may unintentionally place sensitive data in inappropriate AI systems.

Data Tiers

- Public information that can be used in approved public or enterprise tools.
- Internal information that should stay within approved organizational environments.
- Confidential or sensitive operational data that require restricted access and review.
- Protected health information or legally protected records that require HIPAA, security, contractual, and minimum-necessary controls.
- Tribal or sovereign data, when applicable, that require governance by the appropriate Tribal authority.

Operational Rules

- Define which AI tools are approved for each data tier.
- Require data use agreements, business associate agreements, or vendor addenda when needed.
- Prohibit restricted information in public consumer AI tools.
- Document data sources, retention, access, and model-training restrictions.

Implementation Steps

- Train staff using real examples from their workflows.
- Include tiered data review in intake, procurement, governance, and incident response.
- Review vendor claims about data retention, training, logging, and deletion.

Why This Matters for Your Practice

Clear data tiers turn privacy and security rules into daily practice. Staff need to know which data can be used in which AI environments, what approvals are needed, and what tools are prohibited for sensitive information.

Reflection Questions

- What data tiers does the department use?
- Which AI tools are approved for each tier?
- What vendor, contract, logging, retention, and training restrictions are required?

Practical Exercise

Create a table that maps public, internal, confidential, protected, restricted, and sovereign data categories to approved AI environments and required review steps.

Expected Takeaways

- Tiered data classification table
- Approved AI environment list
- Data-use approval pathway

Purpose

Tiered data use gives staff clear rules for which data can be used with which AI tools. It prevents accidental disclosure and supports consistent decisions across programs.

Typical Tiers

A practical framework may classify data as public, internal, confidential, or highly restricted. Highly restricted data may require in-house systems, approved enterprise tools, specific vendor terms, or complete prohibition from AI use.

AI environment decisions

- Public AI tools may be limited to public or fully de-identified content.
- Enterprise AI tools may support internal drafting if contract and security terms are approved.
- In-house or RAG systems may be needed for confidential reference material.

- Highly restricted data require explicit privacy, legal, security, and governance approval.
- Tribal data requires governance by the appropriate tribal authority.

References and Resources

- HIPAA. Privacy requirements when protected health information may be involved. https://www.hhs.gov/hipaa/index.html
- FedRAMP. Cloud security authorization context for systems handling sensitive government data. https://www.fedramp.gov/
- NIH Tribal Health Research Office. Tribal data governance and tribal sovereignty considerations. https://dpcpsi.nih.gov/thro

Environmental and Resource Impacts

AI can carry energy, water, hardware, e-waste, procurement, and environmental justice implications that should be considered before scale.

AI systems can carry resource impacts related to compute, energy, water, hardware, infrastructure, and cost. These issues are not always visible to program teams, but they matter for responsible public sector decision-making and environmental justice.

Not every AI use requires the same level of review. A small drafting task has different implications than a high-volume, always-on, large-model deployment. Departments should ask whether the AI approach is proportionate to the public health value.

Your training should help staff compare alternatives. Sometimes a smaller model, retrieval-based system, existing analytic method, or non-AI workflow change may achieve the goal with less cost and burden.

Module Overview

AI systems can require substantial compute, energy, water, hardware, and vendor infrastructure. Public health departments do not need to perform complex environmental modeling for every use, but they should consider whether a proposed AI use is proportionate to the public health value, whether lower-impact alternatives exist, and whether infrastructure burdens raise environmental justice concerns.

Review Questions

- Is the AI use necessary, or could a simpler analytic, workflow, or rule-based approach achieve the same goal?
- Will the vendor disclose sustainability practices, data center location, energy use, water use, or environmental commitments?
- Does the project require large-scale model training, high-volume generation, or continuous processing?
- Could infrastructure siting or resource burden affect communities already experiencing environmental inequities?

Procurement Relevance

- Ask vendors for sustainability, compute, hosting, and data center information when relevant.
- Compare lower-impact model sizes, retrieval approaches, or existing enterprise tools.
- Include resource impacts in funding and sustainment planning.

Decision Standard

- Higher resource burden should be justified by clear public health value, governance approval, and feasible sustainment.
- Environmental review should be documented for high-compute or large-scale deployments.

Why This Matters for Your Practice

AI systems can have energy, water, hardware, cost, and environmental justice implications. Departments should consider whether the public health value of a high-compute use justifies its resource burden and whether lower-impact alternatives exist.

Reflection Questions

- Is AI necessary for this task, or would a simpler approach work?
- What sustainability and infrastructure information should vendors disclose?
- Could resource burdens affect communities already facing environmental inequities?

Practical Exercise

Review one proposed AI project and compare a high-compute option, a smaller model or RAG option, and a non-AI option. Document benefits, limitations, costs, and resource implications.

Expected Takeaways

- Environmental and resource impact review
- Lower-impact alternatives comparison
- Procurement sustainability questions

Why It Belongs in Public Health

AI systems can consume substantial energy, water, hardware, and cloud resources. These impacts are public health concerns when they intersect with environmental justice, climate resilience, procurement, and community burden.

Practical Review

Most agency reviews can be lightweight: identify whether the use is compute-intensive or vendor-hosted, ask about sustainability and data center practices, consider lower-impact alternatives, and document whether the public health value justifies the resource burden.

Review factors

- Energy and water intensity of the AI service.
- Hardware, e-waste, and cloud infrastructure implications.
- Data center location and environmental justice concerns.
- Vendor sustainability commitments and reporting.
- Lower-impact alternatives such as smaller models, retrieval systems, or workflow redesign.

References and Resources

- ASPPH AI Framework Report. Resource, public health systems, and responsible AI framing. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- EPA Environmental Justice. Environmental justice context for reviewing infrastructure, procurement, and community burden. https://www.epa.gov/environmentaljustice
- NIST AI Risk Management Framework. Risk management structure for documenting environmental and operational tradeoffs. https://www.nist.gov/itl/ai-risk-management-framework

The AI Bridge Professional Role

The bridge professional translates across public health programs, IT, legal/privacy, procurement, vendors, leadership, data teams, and community partners.

The AI bridge professional is a role or function that helps departments translate across disciplines. Public health AI work often stalls when program needs, technical requirements, legal concerns, procurement rules, and community expectations are discussed separately.

Bridge capacity can be a dedicated position, a shared role, or a team function. What matters is that someone is responsible for connecting workflows, governance materials, data constraints, vendor claims, staff training, and implementation realities.

Your training should help departments identify where bridge work is needed. Smaller agencies may need regional support, shared services, or designated staff with protected time rather than a new full-time position.

Module Overview

The AI bridge professional is a role or function that connects public health programs, data teams, IT, privacy, legal, procurement, vendors, leadership, and community partners. This role helps translate public health needs into technical requirements and technical risks into practical governance and implementation decisions.

Core Responsibilities

- Help programs define use cases, workflow needs, success measures, and documentation requirements.
- Coordinate governance submissions, risk review, stakeholder engagement, and procurement questions.
- Translate model limitations, data quality issues, validation findings, and vendor claims for nontechnical audiences.
- Support staff training, change management, and implementation troubleshooting.

Where the Role Fits

- During readiness assessment to identify capacity gaps.
- During governance to prepare complete intake and review materials.
- During prioritization to compare public health value, feasibility, and risk.
- During deployment to coordinate training, monitoring, and incident escalation.

Staffing Models

- A large department may create a dedicated role or small AI implementation office.
- A smaller department may assign bridge responsibilities to an informatics, quality improvement, data modernization, or program leader with protected time.
- Regional or shared-service models may support multiple jurisdictions.

Why This Matters for Your Practice

Many AI problems are translation problems: program needs do not automatically become technical requirements, and technical risks do not automatically become operational controls. Bridge professionals help keep AI implementation coherent across teams.

Reflection Questions

- Who can translate between program, data, IT, privacy, procurement, legal, equity, communications, vendors, and leadership?
- What authority and protected time does the bridge role need?
- How will bridge work be sustained in smaller departments?

Practical Exercise

Define the bridge role for one AI project. Identify responsibilities during intake, governance review, procurement, implementation, staff training, monitoring, and issue escalation.

Expected Takeaways

- Bridge professional role description
- Cross-functional coordination map
- Implementation support responsibilities

Role Definition

The AI bridge professional is a public health translator: someone with enough technical understanding to work with IT and vendors, enough program knowledge to understand workflows, and enough communication skill to support leadership, governance, and community conversations.

Why Agencies Need This Role

AI projects often fail at handoffs between program goals, data realities, legal constraints, vendor claims, staff concerns, and community expectations. A bridge role helps turn those handoffs into documented decisions.

Common responsibilities

- Translate use case needs into governance and technical requirements.
- Help staff document workflows, risks, data needs, and human review points.
- Coordinate between program owners, IT, legal/privacy, procurement, equity, and vendors.
- Support AI literacy, change management, and community-facing explanations.
- Track unresolved questions and route them to the right decision-makers.

References and Resources

- ASPPH AI Framework Report. Bridge professional concept and AI workforce implications. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/

- Public Health Foundation TRAIN Learning Network. Training infrastructure for building cross-functional workforce capacity. https://www.train.org/
- CDC STLT Data Connection. Peer-learning context for public health data modernization and implementation support. https://www.cdc.gov/data-modernization/php/stlt-data-connection/index.html

Federal Funding Opportunities

AI work can often be positioned as a capability layer on top of data modernization, infrastructure, workforce, and equity investments.

Funding strategy should make the full cost of responsible AI visible. Software is only one cost. Departments may also need data modernization, integration, security review, staff training, governance time, evaluation, monitoring, accessibility support, translation, and community engagement.

AI work can often align with existing public health investment priorities such as data modernization, workforce capacity, emergency preparedness, health equity, communications, and service delivery. The key is to connect the AI use case to a real public health capability need.

Your training should prepare staff to avoid unfunded pilots. A responsible funding plan explains what is needed for planning, pilot, scale, and sustainment, and what happens when short-term funding ends.

Module Overview

AI projects need funding for more than software licenses. Responsible implementation may require data modernization, cloud or secure infrastructure, staff time, training, governance, legal and procurement review, evaluation, monitoring, translation, accessibility, community engagement, and sustainment. Funding strategy should be developed before pilots are approved for scale.

Funding Alignment

- Position AI as an enabling capability within data modernization, workforce, equity, emergency preparedness, surveillance, communications, and service delivery investments.
- Match use cases to allowable costs in DMI, PHIG, HRSA, FEMA, EPA, SAMHSA, NIH, NSF, or state and local funding streams when appropriate.
- Avoid creating pilots that cannot be maintained after short-term funds end.

Budget Categories

- Technology licenses, hosting, security review, integration, and vendor support.
- Staff time for governance, implementation, training, monitoring, and evaluation.
- Community engagement, accessibility, translation, and communications support.
- Evaluation, model monitoring, incident response, and policy maintenance.

Sustainment Questions

- Who pays after the pilot?
- What staffing is required to keep the system safe and useful?
- What costs increase if the system scales across programs or jurisdictions?
- What functions can be shared regionally or through partnerships?

Why This Matters for Your Practice

AI projects often require resources that are easy to overlook: staff time, governance, training, validation, data modernization, monitoring, accessibility, community engagement, and sustainment. Funding strategy should make the full cost of responsible implementation visible.

Reflection Questions

- What costs are required before pilot, during deployment, and after scale?
- Which funding streams align with data, workforce, equity, emergency response, surveillance, or service delivery goals?
- What happens when short-term funding ends?

Practical Exercise

Build a funding map for one approved use case that includes pilot costs, scale costs, sustainment costs, staffing, evaluation, governance, and possible funding sources.

Expected Takeaways

- Funding alignment table
- Sustainment plan
- Budget categories for responsible AI implementation

Funding Strategy

Public health departments do not need to fund AI from scratch. AI can often be positioned as a capability layer on top of data modernization, workforce, infrastructure, equity, and evaluation investments.

DMI and PHIG

CDC Data Modernization Initiative investments can support interoperable, cloud-ready, standards-based infrastructure. Public Health Infrastructure Grant resources can support workforce, data systems, organizational capacity, and health equity work needed for responsible AI.

Reference Table

- CDC Data Modernization Initiative - Shared data infrastructure, cloud environments, APIs, standards, and AI-ready data pipelines.
- Public Health Infrastructure Grant - AI-related workforce, governance capacity, data systems, organizational change, and equity monitoring.
- HRSA digital health and telehealth programs - Patient communication drafting, document search, and health equity analysis.
- NIH / NSF smart health programs - Research partnerships, model evaluation, fine-tuning, and agentic system development.
- SAMHSA technology-based services - Early identification, intervention support, client communication, and care coordination.
- EPA environmental justice grants - GeoAI, environmental health analysis, and heat risk mapping.
- FEMA emergency management grants - Disaster resilience and agentic outbreak forecasting workflows.

References and Resources

- CDC Data Modernization Initiative. Potential alignment for data infrastructure and AI-ready modernization investments. https://www.cdc.gov/surveillance/data-modernization/index.html
- Public Health Infrastructure Grant. Potential alignment for workforce, data systems, organizational capacity, and equity investments. https://www.cdc.gov/infrastructure-phig/php/index.html
- HRSA Telehealth Programs. Potential alignment for digital health, telehealth, and access-related AI use cases. https://www.hrsa.gov/rural-health/telehealth

Procurement and Vendor Oversight

AI procurement requires public health departments to evaluate vendor claims, contract terms, data rights, model documentation, auditability, accessibility, cybersecurity, support, and exit plans before an AI-enabled product is approved.

Vendor oversight is a public health governance issue, not only a purchasing task. A product may look efficient in a demonstration but still create risks if the department cannot verify how data are used, how outputs are produced, how errors are handled, or how the system can be audited.

You should be trained to ask practical questions before procurement moves forward. What decision or workflow will the tool affect? What data are involved? Who owns the output? What performance evidence exists for populations similar to the department's communities? What contract terms protect the department if the system fails?

This module prepares you to use procurement as a safeguard. Strong procurement requirements can turn responsible AI principles into enforceable vendor obligations.

Module Overview

Public health departments often encounter AI through vendor platforms, not custom-built internal systems. Procurement staff, program owners, IT, security, privacy, legal, and governance members need shared criteria for evaluating vendor claims before contracts are signed or pilots begin.

What Staff Should Look For

- Clear description of what AI capability is included and whether it is generative, predictive, agentic, or rules-based.
- Model documentation, validation evidence, limitations, monitoring approach, and known failure modes.
- Contract terms covering data ownership, data reuse, model training, audit rights, breach notification, accessibility, support, subcontractors, and termination.
- Evidence that the vendor can support public-sector privacy, security, records, and procurement obligations.

Public Health Procurement Questions

- What data will the vendor receive, store, process, or use to improve models?
- Can the department audit performance, errors, bias, access logs, and system changes?
- What happens if the model changes, performance declines, guidance changes, or the contract ends?
- How will the vendor support language access, disability access, incident response, and staff training?

Implementation Check

- Route vendor AI proposals through governance before pilot approval.
- Use risk tiering to decide what validation, privacy, legal, and procurement review is required.
- Do not accept vendor performance claims without evidence, test data, monitoring plans, and department review.

Why This Matters for Your Practice

Most departments will buy or inherit AI-enabled systems before they build their own. Vendor oversight protects public health authority, data rights, security, equity, and operational continuity.

Reflection Questions

- What AI-enabled vendor tools are already in use or under consideration?
- Which contract terms would prevent the department from auditing or safely exiting the system?
- Who must review vendor claims before a pilot or purchase is approved?

Practical Exercise

Review a sample AI vendor proposal and identify missing documentation, contract protections, validation evidence, data-use terms, accessibility requirements, and governance questions.

Expected Takeaways

- Vendor AI review checklist
- Procurement questions for governance
- Draft AI contract requirements

References and Resources

- NIST AI Risk Management Framework. Use to structure vendor risk, documentation, transparency, and accountability review. https://www.nist.gov/itl/ai-risk-management-framework
- FedRAMP. Cloud security authorization context for vendor-hosted systems that may handle sensitive government data. https://www.fedramp.gov/
- HHS Civil Rights and AI. Equity, nondiscrimination, and civil rights considerations for AI-enabled vendor tools. https://www.hhs.gov/civil-rights/for-individuals/special-topics/ai/index.html

Privacy, Confidentiality, and Public Records

AI use in public health must account for PHI, PII, confidential public health records, investigation data, data-sharing limits, retention rules, and public records obligations.

Privacy training needs to be practical. You should not have to interpret complex law in the moment of writing a prompt. They need clear categories, approved tools, examples of prohibited data, and a simple escalation path when they are unsure.

Public records issues are especially important for government agencies. AI-assisted drafts, prompts, summaries, and decision notes may become part of the administrative record. Departments should decide how these materials are stored and reviewed before AI use becomes routine.

The goal is not to stop all AI use. The goal is to match the sensitivity of the data to the correct environment, review pathway, and documentation standard.

Module Overview

AI can change where data travel, who can access them, how long they are retained, and whether outputs become records. Staff need plain-language training on what information may be used, which AI environments are approved, and when legal, privacy, records, or data-sharing review is required.

Key Data Categories

- Public information that can safely be used in approved public-facing contexts.
- Internal information that may be used in enterprise tools under agency policy.
- Confidential, identifiable, or protected information that requires restricted environments and review.
- Highly restricted data such as investigations, legal matters, sensitive surveillance data, or data governed by special agreements.

Public Records and Retention

- AI prompts, outputs, summaries, edits, decisions, and approvals may become records depending on jurisdiction and use.
- Departments should define when AI-assisted work must be retained, where it is stored, and how it is labeled.
- You should avoid creating unofficial records in personal accounts or unapproved AI tools.

Implementation Check

- Classify data before selecting an AI tool.
- Review data use agreements, vendor terms, retention rules, and public records obligations.
- Train staff on prohibited data entry and escalation pathways for uncertain cases.

Why This Matters for Your Practice

Privacy, confidentiality, and records controls are foundational for STLT public health AI because departments handle sensitive information, regulatory records, investigations, partner data, and public accountability obligations.

Reflection Questions

- Which types of department data should never be entered into public AI tools?
- Where should AI prompts and outputs be stored when they support official work?
- Who decides whether a proposed AI use requires privacy, legal, or records review?

Practical Exercise

Sort example AI tasks into public, internal, confidential, and highly restricted data-use categories, then identify the approved AI environment and review pathway for each.

Expected Takeaways

- AI data classification examples
- Privacy escalation rules
- Public records documentation notes

References and Resources

- HIPAA. Federal privacy framework when protected health information may be involved. https://www.hhs.gov/hipaa/index.html
- CDC Public Health Data Policy and Standards. Public health data policy and standards context for responsible data use. https://www.cdc.gov/surveillance/data-modernization/policy.html
- NIST Privacy Framework. Privacy risk management structure for data processing and technology decisions. https://www.nist.gov/privacy-framework

Model Validation and Evaluation Basics

Validation helps departments decide whether an AI system performs well enough, for the right users and populations, before it is piloted, scaled, or sustained.

Validation is not just a technical score. A technically impressive model may be inappropriate if it creates staff burden, performs poorly for certain communities, or cannot be explained well enough for public health accountability.

You should understand the difference between vendor validation, local validation, pilot evaluation, and ongoing monitoring. Vendor evidence may be useful, but the department still needs to know whether the system works in its own data, workflows, populations, and operating conditions.

This module should help you ask better questions of data scientists and vendors, and help governance groups interpret validation evidence before decisions are made.

Module Overview

Validation is the disciplined process of checking whether an AI-supported workflow is accurate, reliable, safe, equitable, usable, and appropriate for its intended public health purpose. It should be understandable to nontechnical staff because validation results inform governance decisions.

What Validation Should Cover

- Purpose fit: whether the system solves the public health problem it is intended to support.
- Performance: accuracy, false positives, false negatives, completeness, reliability, and stability.
- Equity: subgroup performance, missing voices, language access, disability access, and geographic variation.
- Workflow impact: staff time, burden, usability, review quality, and downstream consequences.
- Safety: harmful outputs, hallucinations, unsafe recommendations, and incident triggers.

Evaluation Over Time

- Pilot evaluation should compare the AI-supported workflow to the current process.
- Monitoring should continue after deployment because models, data, staff behavior, guidance, and populations change.
- Governance should define thresholds for continue, revise, pause, scale, or retire decisions.

Implementation Check

- Define acceptance criteria before testing begins.
- Use realistic examples and local workflow conditions.
- Document limitations and do not scale until governance reviews evaluation evidence.

Why This Matters for Your Practice

Validation protects agencies from scaling systems that are inaccurate, inequitable, poorly integrated, or untrusted by staff and communities.

Reflection Questions

- What would count as acceptable performance for this workflow?
- Which groups or locations need subgroup review?
- What evidence should governance require before scale-up?

Practical Exercise

Design a validation plan for one proposed AI use case, including test examples, performance measures, subgroup checks, human review expectations, and go/no-go criteria.

Expected Takeaways

- Validation plan outline
- Evaluation metric list
- Governance decision thresholds

References and Resources

- NIST AI Risk Management Framework. Risk and measurement framework for validation, monitoring, and governance. https://www.nist.gov/itl/ai-risk-management-framework
- WHO: Regulatory considerations on artificial intelligence for health. Health AI validation, safety, transparency, and monitoring considerations. https://iris.who.int/handle/10665/373421
- HHS Civil Rights and AI. Equity and nondiscrimination considerations when AI affects health and human services. https://www.hhs.gov/civil-rights/for-individuals/special-topics/ai/index.html

AI Incident Response

AI incident response defines how staff report, triage, investigate, correct, communicate, and document problems involving AI-supported systems or outputs.

AI incident response should feel familiar to public health departments because it builds on existing approaches to privacy, security, quality improvement, emergency response, and corrective action. What is new is the range of AI-specific failure modes that you may not recognize without training.

The most important training message is simple: if an AI output seems wrong, harmful, biased, unsafe, or outside approved use, staff should know where to report it and should not have to resolve it alone.

Incident response also supports learning. Every incident should feed back into policy updates, tool design, staff training, monitoring, and vendor oversight.

Module Overview

AI incidents can involve incorrect outputs, privacy exposure, biased results, unsafe recommendations, unauthorized use, vendor failure, staff over-reliance, public misinformation, or model performance drift. Staff need to know what to report and what happens next.

Incident Types

- Harmful or inaccurate output used in a public health workflow.
- Sensitive data entered into an unapproved AI tool or exposed through a vendor system.
- Bias or disparate impact concern raised by staff, partners, or community members.
- Model drift, hallucination, unsafe recommendation, or unexplained performance change.
- Public-facing AI response that creates confusion, misinformation, or loss of trust.

Response Steps

- Capture the issue, system, date, user, output, data involved, and immediate risk.
- Triage severity and determine whether the system should continue, be limited, paused, or shut off.
- Notify privacy, security, legal, program leadership, communications, vendor, or governance contacts as appropriate.
- Document corrective actions, user communications, policy updates, and evidence of closure.

Implementation Check

- Train staff to report concerns without blame.
- Define emergency contacts and pause authority before deployment.
- Review incidents regularly through governance and continuous improvement.

Why This Matters for Your Practice

AI incidents can affect privacy, public trust, legal compliance, staff confidence, equity, and operational safety. A clear response pathway reduces harm and improves governance.

Reflection Questions

- What AI-related problems should staff be required to report?
- Who has authority to pause an AI-supported workflow?

- How will lessons from incidents update training, policy, and monitoring?

Practical Exercise

Walk through a tabletop scenario involving an incorrect AI-generated public message or privacy exposure, then document reporting steps, escalation contacts, communications needs, corrective actions, and governance review.

Expected Takeaways

- AI incident scenario response
- Escalation pathway
- Corrective action log

References and Resources

- NIST AI Risk Management Framework. Governance and response structure for AI-related harms and failures. https://www.nist.gov/itl/ai-risk-management-framework
- WHO: Ethics and governance of artificial intelligence for health. Health AI accountability, safety, and oversight principles. https://www.who.int/publications/i/item/9789240029200
- HHS Office for Civil Rights Breach Notification. Context for privacy breach response when protected health information may be involved. https://www.hhs.gov/hipaa/for-professionals/breach-notification/index.html

Role-Based AI Use Guidelines

Different public health roles need different AI permissions, responsibilities, training expectations, review duties, and escalation pathways.

AI policy becomes much easier to follow when you can see how it applies to their actual role. A communications specialist, epidemiologist, procurement officer, supervisor, and community engagement lead should not receive identical instructions and be expected to infer the rest.

Role-based guidance also supports security and accountability. You should only have access to the AI tools and data environments they need, and they should understand what documentation is required when they use AI for official work.

This module helps you move your department from broad AI literacy into operational readiness.

Module Overview

Role-based AI guidelines translate policy into daily practice. Leaders, program colleagues, analysts, communications staff, IT/security, privacy/legal, procurement, equity staff, supervisors, and governance members each need practical guidance for their responsibilities.

Role Differences

- Leaders approve direction, resources, governance authority, and risk tolerance.
- Program colleagues define workflow needs, review outputs, and report concerns.
- Analysts and informatics staff support data quality, validation, monitoring, and documentation.
- Communications staff review public-facing language, plain language, translations, and misinformation risk.
- Legal, privacy, security, procurement, and equity staff apply specialized review requirements.

Training Expectations

- Every role needs baseline AI literacy, prohibited-use guidance, data-use rules, and escalation expectations.
- High-impact roles need deeper training on governance, validation, procurement, public communication, or monitoring.
- Supervisors need guidance on approving use, checking documentation, and supporting staff adoption.

Implementation Check

- Create a role-by-role matrix of allowed uses, prohibited uses, required approvals, and documentation expectations.
- Connect role guidance to onboarding, annual training, governance participation, and tool access.

Why This Matters for Your Practice

Role clarity prevents inconsistent AI use, unsupported individual experimentation, unreviewed public communications, and unclear accountability.

Reflection Questions

- Which roles need permission to use which AI tools?
- Which roles must review outputs before they are used?
- Which roles need advanced training before governance or procurement decisions?

Practical Exercise

Build a role-based AI use matrix for one department, including allowed uses, prohibited uses, approval requirements, documentation, and escalation contacts.

Expected Takeaways

- Role-based AI use matrix
- Training assignment plan
- Access and approval rules

References and Resources

- ASPPH AI Framework Report. Public health workforce, AI literacy, and role preparation framing. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- CDC Public Health Workforce Development. Public health workforce capability-building context. https://www.cdc.gov/workforce/
- Public Health Foundation TRAIN Learning Network. Training platform for role-based public health workforce development. https://www.train.org/

Responsible Prompting and Review

Responsible prompting helps staff use generative AI safely by protecting data, asking precise questions, requiring sources when needed, and reviewing outputs before use.

Responsible prompting training should be hands-on. Staff learn best by comparing a vague prompt, a safer prompt, and a reviewed final output. The point is not to create perfect prompts; the point is to create safer drafts and stronger review habits.

Departments should be clear that generative AI can support drafting and synthesis, but it does not replace subject matter expertise, legal review, privacy review, or communications approval.

This module should give staff enough confidence to use approved tools productively while recognizing when a task is too sensitive, too consequential, or too uncertain for AI assistance.

Module Overview

Staff need practical guidance for everyday generative AI use. Prompting is not magic language; it is a structured way to define task, audience, source material, constraints, review criteria, and prohibited data. The review step is more important than the prompt.

Prompting Practices

- State the task, audience, purpose, tone, and output format.
- Provide approved source material when the output must reflect agency policy or current guidance.
- Ask the tool to identify assumptions, limitations, uncertainty, and items needing human review.
- Never include PHI, confidential records, investigation details, credentials, or restricted partner data in public AI tools.

Review Practices

- Check factual claims, dates, citations, calculations, recommendations, and legal or policy statements.
- Review plain language, accessibility, cultural context, language access, and potential stigma.
- Confirm that the output does not imply an action, authority, or guarantee the department cannot support.
- Document AI assistance when required by agency policy.

Implementation Check

- Train staff with examples from their own workflows.
- Provide approved prompt templates and review checklists.
- Use escalation rules for public-facing, high-risk, or uncertain outputs.

Why This Matters for Your Practice

Responsible prompting and review reduce the chance that staff use AI in ways that expose data, spread misinformation, create inaccessible communication, or bypass expert review.

Reflection Questions

- Which AI-assisted drafting tasks should be allowed for staff?
- Which outputs require source verification, supervisor review, or communications approval?
- How should staff document AI-assisted work?

Practical Exercise

Rewrite three unsafe prompts into responsible prompts, then review sample AI outputs for data exposure, hallucinations, tone, equity concerns, missing sources, and required approvals.

Expected Takeaways

- Responsible prompt examples
- Output review checklist
- Documentation expectations

References and Resources

- OpenAI Prompt Engineering Guide. Core prompt engineering practices for clear instructions, examples, task decomposition, and systematic iteration. https://platform.openai.com/docs/guides/prompt-engineering
- Google Cloud Prompt Engineering Guide. High-quality overview of prompt components, prompt patterns, and how prompts shape generative AI outputs. https://cloud.google.com/discover/what-is-prompt-engineering
- Microsoft Prompt Engineering Techniques. Operational prompting techniques for structured outputs, grounding, examples, and model-specific prompt refinement. https://learn.microsoft.com/en-us/azure/ai-services/openai/concepts/prompt-engineering
- NIST Generative AI Profile. Risk management concepts for generative AI systems and outputs. https://www.nist.gov/itl/ai-risk-management-framework
- WHO: Ethics and governance of artificial intelligence for health. Ethics and governance principles for health AI use. https://www.who.int/publications/i/item/9789240029200
- CDC Clear Communication Index. Useful for reviewing plain-language public health materials. https://www.cdc.gov/ccindex/

Community Engagement for AI

Community engagement helps departments understand how AI-supported workflows may affect trust, access, surveillance concerns, language needs, disability access, and public accountability.

Community engagement is not a public relations step after technical decisions are complete. It is a way to improve design, identify harms, clarify accountability, and build trust before implementation choices harden.

AI systems may change how people are contacted, prioritized, informed, or served. Communities should be able to understand those changes and raise concerns before the department launches a tool that affects them.

This module should help staff design engagement that is specific enough to influence AI decisions rather than simply announcing that AI will be used.

Module Overview

Community engagement for AI should happen before use cases are finalized, especially when a system affects public communication, surveillance, outreach, service access, prioritization, or resource allocation. Engagement helps departments surface concerns that technical testing may miss.

Engagement Focus Areas

- What the AI-supported workflow will and will not do.
- What data are involved and how privacy is protected.
- What decisions remain human and how concerns can be raised.
- How language, disability, digital access, rural context, Tribal context, immigration concerns, or historical mistrust may affect the workflow.

Engagement Methods

- Listening sessions, advisory groups, community partner consultation, usability testing, plain-language review, and feedback forms.
- Internal engagement with frontline staff who understand how communities experience department services.
- Ongoing feedback after launch, not only pre-implementation input.

Implementation Check

- Document who was engaged, what concerns were raised, what changed, and what remains unresolved.
- Route unresolved community concerns to governance.
- Close the feedback loop with plain-language summaries.

Why This Matters for Your Practice

Engagement helps departments avoid AI uses that appear efficient internally but create confusion, mistrust, stigma, surveillance concerns, or access barriers externally.

Reflection Questions

- Which communities or partners may be affected by the proposed AI use?
- What should be explained in plain language before implementation?
- How will community feedback change governance, design, monitoring, or communication?

Practical Exercise

Create a community engagement plan for one AI-supported workflow, including audiences, engagement methods, plain-language explanation, feedback documentation, and governance escalation.

Expected Takeaways

- AI community engagement plan
- Plain-language AI explanation
- Feedback-to-decision log

References and Resources

- CDC Health Equity. Public health equity context for community engagement and accountability. https://www.cdc.gov/healthequity/
- ASPPH AI Framework Report. Human-centered AI, missing voices, and public accountability framing. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- Blueprint for an AI Bill of Rights. Notice, explanation, human alternatives, and protections from automated harms. https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/

Accessibility and Language Access

AI-supported public health materials, tools, dashboards, chatbots, translations, and workflows should be reviewed for disability access, language access, readability, cultural context, and usability.

AI can make communication faster, but speed is not the same as access. Public health departments need to know whether messages can be understood, whether tools can be used by people with disabilities, and whether translations are accurate enough for the purpose.

Language access and accessibility are also trust issues. Communities may lose confidence if AI-generated messages are confusing, culturally inappropriate, inaccessible, or inconsistent with staff communication.

This module helps you build access review into AI planning rather than treating it as a final proofreading step.

Module Overview

AI can support translation, plain-language rewriting, alternative formats, and communication drafts, but those outputs still need human review. Accessibility and language access should be part of planning, testing, communications approval, and monitoring.

What Needs Review

- AI-generated public health messages, FAQs, chatbot responses, translations, summaries, forms, dashboards, and outreach lists.
- Reading level, clarity, disability access, screen-reader compatibility, captions, plain-language structure, cultural context, and stigmatizing language.
- Whether non-digital and human assistance pathways remain available.

Translation and Language Access

- AI translation should not be used for high-stakes public messaging without qualified human review.
- Departments should review dialect, cultural context, health terminology, and risk communication accuracy.
- You should document which languages were reviewed and by whom.

Implementation Check

- Include accessibility and language access in use case scoring.
- Test public-facing tools with users before launch.
- Monitor complaints, failed interactions, correction requests, and access barriers after deployment.

Why This Matters for Your Practice

Accessible and language-appropriate AI-supported communication is essential for equitable public health service delivery and public trust.

Reflection Questions

- Which AI-supported outputs will be public-facing or client-facing?
- Who reviews translations, accessibility, readability, and cultural context?
- What non-AI or human pathway remains available?

Practical Exercise

Review an AI-generated public health message and identify changes needed for plain language, accessibility, language access, cultural context, and human assistance pathways.

Expected Takeaways

- Accessibility and language access review notes
- Revised public message
- Monitoring indicators for access barriers

References and Resources

- HHS Section 1557. Nondiscrimination and language access context in health programs and activities. https://www.hhs.gov/civil-rights/for-individuals/section-1557/index.html
- ADA.gov. Federal accessibility requirements and guidance. https://www.ada.gov/
- CDC Clear Communication Index. Plain-language review support for health communication materials. https://www.cdc.gov/ccindex/

Cybersecurity and Secure AI Use

Secure AI use requires approved environments, access controls, vendor security review, audit logs, prompt-injection awareness, data-loss prevention, and incident escalation.

AI tools can blur the boundary between software, data processing, content generation, and workflow automation. That makes security review especially important before departments connect AI to email, document repositories, case systems, dashboards, or public-facing services.

Your training should make secure AI use concrete. Which tools are approved? What data are prohibited? What are signs of prompt injection or suspicious AI behavior? Who should be notified when something goes wrong?

This module helps you understand why security is a shared responsibility and why shortcuts can create agency-wide risk.

Module Overview

AI security is broader than protecting the model. It includes user accounts, data flows, integrations, APIs, vendor hosting, prompts, outputs, logs, plugins, browser extensions, and connections to existing systems. Staff need to know how secure AI use differs from general software use.

Security Concerns

- Sensitive data entered into unapproved tools or retained by vendors.
- Prompt injection or malicious content that manipulates AI behavior.
- Unauthorized access through weak role controls, shared accounts, or unmanaged integrations.
- Unverified AI-generated code, formulas, or instructions introduced into operational systems.
- Vendor outages, model changes, or compromised services affecting public health operations.

Secure Practices

- Use only approved AI tools for official work.
- Apply role-based access, multi-factor authentication, logging, data-loss prevention, and least-privilege access.
- Review vendor security documentation, hosting, subcontractors, breach notification, and audit rights.
- Train staff to treat AI outputs and external content as potentially unsafe until reviewed.

Implementation Check

- Include security review before pilot approval.
- Test integrations in controlled environments.
- Coordinate AI incident response with cybersecurity and privacy response plans.

Why This Matters for Your Practice

Cybersecurity protects public health operations, sensitive data, system integrity, and community trust.

Reflection Questions

- Which AI tools are approved for official work?
- Which integrations or data flows require security review?
- How should staff report suspicious AI behavior, data leakage, or unauthorized access?

Practical Exercise

Map the security controls needed for one AI use case, including data classification, access controls, vendor hosting, integrations, logging, incident response, and staff training.

Expected Takeaways

- AI security review notes
- Approved-use rules
- Security escalation checklist

References and Resources

- NIST Cybersecurity Framework. Cybersecurity risk management structure for organizational systems. https://www.nist.gov/cyberframework

- CISA AI and Cybersecurity. Cybersecurity context for AI systems and AI-enabled threats. https://www.cisa.gov/ai
- FedRAMP. Cloud security authorization context for vendor-hosted government systems. https://www.fedramp.gov/
- OWASP LLM01: Prompt Injection. Security guidance on prompt injection risks, including malicious instructions, indirect prompt injection, and mitigation considerations for LLM applications. https://genai.owasp.org/llmrisk/llm01-prompt-injection/

When Not to Use AI

Departments need a clear decision framework for situations where AI is inappropriate, premature, too risky, unsupported by data, legally uncertain, or not worth the operational burden.

You may feel pressure to find AI uses because the technology is highly visible. Your training should make it acceptable to conclude that AI is not the right tool for a particular problem.

The strongest AI programs are not the ones with the most pilots. They are the ones that select use cases carefully, say no to poor fits, and invest in readiness before moving forward.

This module gives you a shared language for declining, deferring, or redesigning proposals without making the discussion feel anti-innovation.

Module Overview

Responsible AI adoption includes saying no. Some tasks should remain human-only, some require more readiness work before AI is considered, and some can be improved more safely through workflow redesign, staffing, training, or better data systems.

Red Flags

- The use case affects rights, enforcement, eligibility, resource allocation, public warnings, or clinical/care decisions without strong governance and human review.
- Data are poor quality, incomplete, unrepresentative, sensitive, or not legally available for the proposed use.
- The department cannot explain, validate, monitor, audit, or pause the system.
- Affected communities or staff have not been engaged.
- The expected benefit does not justify cost, risk, environmental burden, or staff disruption.

Alternative Actions

- Improve the workflow without AI.
- Strengthen data quality, documentation, staffing, or training first.
- Use a simpler rules-based tool, checklist, dashboard, or template.
- Limit AI to internal drafting or evidence organization rather than decision support.
- Pause the proposal until governance conditions are met.

Implementation Check

- Create prohibited-use and restricted-use categories.
- Require governance review for high-impact proposals.
- Document why a use case was rejected, deferred, limited, or redesigned.

Why This Matters for Your Practice

A clear no-go framework protects departments from avoidable harm, unfunded pilots, weak accountability, and loss of trust.

Reflection Questions

- Which public health decisions should remain human-only?
- What conditions would cause governance to reject or defer an AI proposal?
- What non-AI alternatives could solve the problem more safely?

Practical Exercise

Review three proposed AI use cases and decide whether each should proceed, be limited, be deferred, be redesigned without AI, or be rejected.

Expected Takeaways

- AI no-go decision notes
- Restricted-use criteria
- Alternative solution options

References and Resources

- Blueprint for an AI Bill of Rights. Principles for human alternatives, notice, and protection from automated harms. https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/
- NIST AI Risk Management Framework. Risk-based framing for deciding whether and how AI should be used. https://www.nist.gov/itl/ai-risk-management-framework
- ASPPH AI Framework Report. Public health framing for responsible adoption, governance, and human-centered AI. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/

Areas Where AI Can Support Public Health Activities

Expanded examples showing how AI can support specific public health activities, what safeguards are needed, and which plays and tools help agencies move from idea to responsible implementation.

Why This Matters for Your Practice

Support areas help staff move from abstract AI concepts to concrete public health opportunities. They should be used to identify possible use cases, not to skip readiness, governance, stakeholder engagement, or risk review.

Reflection Questions

- Which support areas align with current public health priorities?
- Which areas are feasible given current data, workforce, governance, and funding?
- Which areas could create equity, privacy, transparency, or public trust concerns?

Practical Exercise

Select three support areas and score each for public health value, feasibility, risk, equity impact, data readiness, staff readiness, and sustainment.

Expected Takeaways

- AI opportunity list
- Use case screening notes
- Prioritized concepts for governance review

Areas Where AI Can Support Public Health Activities

These expanded examples are drawn from the learning modules and organized as practical planning topics for public health agencies. Each area includes potential activities, required safeguards, example concepts, and related plays and tools.

Surveillance, Signal Detection, and Outbreak Intelligence

AI can help staff review high-volume surveillance streams, detect unusual patterns, summarize incoming reports, and prioritize signals for epidemiologist review.

Useful Activities

- Summarize surveillance reports into trend, geography, population, and uncertainty sections.
- Flag anomalies or clusters for human investigation.
- Assemble evidence packets from laboratory, syndromic, emergency department, and partner reports.

- Draft internal situation updates from approved source data.

Required Safeguards

- AI should flag and organize information, not declare outbreaks or issue public alerts.
- Epidemiologists should validate signals and document decisions.
- Data latency, false positives, false negatives, and subgroup effects should be monitored.

Learning examples: Summarize surveillance reports, Route high-priority surveillance signals, Detect anomalies, Synthesize outbreak updates

Related plays: 2, 3, 7, 11, 13. Related tools: 7, 13, 14, 40, 43, 46.

Evidence Review, Policy Scans, and Research Synthesis

AI can accelerate first-pass reviews of literature, public health guidance, policy requirements, grant notices, and program examples, while staff verify sources and applicability.

Useful Activities

- Conduct literature scans for emerging threats or interventions.
- Compare policy requirements across jurisdictions.
- Summarize scientific, legal, or funding documents for leadership review.
- Create annotated evidence tables with source links and open questions.

Required Safeguards

- Require citations and source verification.
- Separate evidence strength from opinion, vendor claims, or early findings.
- Have subject matter experts review conclusions before decisions are made.

Learning examples: Literature scans for emerging threats, Policy scans across jurisdictions, Comparative program analysis, Summarize policy documents with human review

Related plays: 1, 3, 7, 8, 13. Related tools: 2, 4, 31, 47.

Plain-Language Communication and Health Education

Generative AI can help convert technical material into accessible drafts for staff, partners, and public audiences when communications staff and subject matter experts review the final content.

Useful Activities

- Draft health education materials, FAQs, and talking points.
- Translate analytic findings into plain language for briefings or partner updates.
- Adapt content for different literacy levels, languages, channels, and audiences.
- Prepare first drafts of public notices about AI-supported activities.

Required Safeguards

- Review for accuracy, tone, reading level, cultural context, accessibility, and current guidance.
- Avoid unsupervised public-facing generation.
- Use approved source material and document human review.

Learning examples: Draft health education materials, Translate analytic findings into plain language, Community-facing FAQs, Public AI use notices

Related plays: 1, 3, 4, 10, 12. Related tools: 19, 20, 28, 45.

Leadership Briefings and Decision Support

AI can organize evidence, options, risks, tradeoffs, and recommended next steps so leaders receive clearer decision packages.

Useful Activities

- Prepare decision briefs from approved evidence and agency records.
- Summarize options, benefits, risks, constraints, and open questions.
- Convert assessment findings into recommended plays, tools, owners, and timelines.
- Draft governance memos for review by accountable staff.

Required Safeguards

- Clearly label AI-assisted drafts.
- Require decision owner sign-off.
- Include uncertainty, limitations, and source references.
- Prevent AI from making final policy, resource, or emergency decisions.

Learning examples: Support decision briefings, Decision owner sign-off, Confidence and limitation notes in briefs

Related plays: 1, 2, 3, 7, 12. Related tools: 1, 6, 7, 13, 45.

Grants, Funding, and Sustainability Planning

AI can help departments align AI work with funding streams, draft grant narratives, compare allowable costs, and prepare sustainment plans.

Useful Activities

- Create first drafts of grant narratives and logic models.
- Map use cases to DMI, PHIG, HRSA, FEMA, EPA, SAMHSA, NIH, NSF, or state/local funding opportunities.
- Estimate staffing, technology, evaluation, training, and sustainment costs.
- Draft budget justification language for reviewed AI projects.

Required Safeguards

- Verify funding requirements, deadlines, eligibility, and allowable costs.
- Avoid committing to AI capabilities that cannot be supported or monitored.
- Include governance, evaluation, accessibility, and change management costs.

Learning examples: Create first drafts of grant narratives, DMI-supported data pipelines, PHIG-supported workforce and governance, Grant-aligned pilots

Related plays: 8, 9. Related tools: 35, 31, 32.

Workflow Routing, Task Queues, and Operational Coordination

Agentic or semi-agentic workflows can help turn incoming information into routed tasks, evidence packets, reminders, and status updates for staff review.

Useful Activities

- Create task queues from incoming reports, meeting notes, complaints, or surveillance items.
- Route materials to the appropriate program, geography, subject matter expert, or governance reviewer.
- Prepare internal reminders for owners, deadlines, approvals, and follow-up actions.
- Update implementation workplans as milestones are completed or blocked.

Required Safeguards

- Define what the AI can create, route, update, or escalate.
- Require audit logs and role-based permissions.
- Give administrators control over assignments, deadlines, and workflow changes.
- Keep humans responsible for final action and external communication.

Learning examples: Create task queues from incoming reports, Coordinate governance submissions, Staff reporting pathways for concerns

Related plays: 3, 6, 10, 11, 12. Related tools: 12, 13, 17, 23, 29, 35.

Equity, Missing Voices, and Community Validation

AI can support equity review by helping teams identify affected groups, compare subgroup performance, draft engagement materials, and track mitigation commitments.

Useful Activities

- Identify underrepresented groups or data gaps during use case review.
- Draft community validation questions and engagement summaries.
- Monitor subgroup performance after deployment.
- Track equity concerns, mitigation owners, and follow-up deadlines.

Required Safeguards

- Use community expertise and affected-user input, not only data analysis.
- Document who was consulted and what changed as a result.
- Do not deploy systems that cannot be evaluated for disparate impact when they affect access, services, enforcement, or public communication.

Learning examples: Identify underrepresented groups, Add community validation before launch, Monitor subgroup performance after deployment, Equity audit before scale

Related plays: 4, 7, 13. Related tools: 3, 20, 22, 46.

Language Access, Accessibility, and Digital Inclusion

AI can assist with translation, plain-language adaptation, accessibility review, and digital service support, but it must be reviewed for accuracy, cultural relevance, and equitable access.

Useful Activities

- Draft translations for human review by qualified language staff or vendors.
- Convert complex guidance into plain-language and accessible formats.
- Evaluate whether AI-supported services create barriers for people without broadband, devices, literacy, English proficiency, or trust in digital systems.
- Develop alternate pathways for people who cannot or do not want to use AI-supported services.

Required Safeguards

- Require human review before public use of translated or accessibility-critical materials.
- Test materials with affected users where feasible.
- Provide non-AI or staff-supported alternatives for important services.

Learning examples: Review AI effects on language access, Evaluate service access impacts, Public AI use notices

Related plays: 1, 4, 7, 10, 13. Related tools: 3, 19, 20, 22, 46.

Privacy, Tiered Data Use, and Secure AI Environments

AI can support work with sensitive information only when departments define data tiers, approved environments, vendor requirements, and review pathways.

Useful Activities

- Classify data as public, internal, confidential, protected, restricted, or sovereign where applicable.
- Match each tier to approved AI tools and required controls.
- Review vendor data retention, model training, logging, deletion, and access claims.

- Prepare data use agreements, BAAs, and procurement addenda when needed.

Required Safeguards

- Restrict PHI and sensitive data from public AI tools.
- Use enterprise, in-house, or controlled RAG environments for restricted materials.
- Document data sources, access rules, and human review requirements.
- Escalate incidents through privacy, security, and governance channels.

Learning examples: Restrict PHI from public AI tools, Approve enterprise AI for internal drafts, Use in-house/RAG systems for restricted materials

Related plays: 2, 3, 11, 12. Related tools: 10, 15, 16, 33, 38.

Procurement, Vendor Review, and Environmental Impact

AI can support procurement analysis, but departments need structured review of vendor transparency, accessibility, auditability, security, data rights, sustainability, and exit requirements.

Useful Activities

- Compare vendor responses against responsible AI, security, privacy, accessibility, and documentation requirements.
- Review environmental and resource impacts for high-compute or scaled systems.
- Draft procurement addenda and contract language for AI-specific obligations.
- Track vendor performance, incidents, and policy updates after deployment.

Required Safeguards

- Do not rely on vendor marketing claims without documentation.
- Require audit rights, data-rights language, model documentation, monitoring obligations, and exit plans.
- Consider lower-impact alternatives where public health value does not justify high compute burden.

Learning examples: Review vendor sustainability claims, Compare lower-impact alternatives, Flag data center environmental justice concerns

Related plays: 3, 7, 8, 9, 12. Related tools: 4, 5, 31, 33, 44.

Workforce Training, AI Literacy, and Bridge Roles

AI can help staff learn and coordinate work, but departments need role-based AI literacy and bridge capacity to connect programs, governance, technology, procurement, and community expectations.

Useful Activities

- Develop role-based training for leaders, program staff, analysts, IT, privacy, procurement, communications, and governance members.
- Create prompt and privacy coaching materials.
- Designate AI champions or bridge professionals to support adoption.
- Track staff concerns, training completion, and workflow barriers.

Required Safeguards

- Do not ask staff to use AI before they understand privacy, hallucination, bias, documentation, and escalation expectations.
- Protect staff time for training and implementation support.
- Use feedback loops to identify workload, trust, and usability problems.

Learning examples: Role-based AI training, Prompt and privacy coaching, Coordinate data modernization with workforce training, Help program staff document AI workflows

Related plays: 5, 6, 10. Related tools: 23, 24, 25, 27, 48, 28, 29.

Performance Improvement, Metrics, and Quality Dashboards

Generative and agentic AI can support public health performance improvement by helping teams define meaningful measures, build baseline profiles, generate improvement ideas, track action plans, and maintain dashboards for ongoing monitoring.

Useful Activities

- Use generative AI to draft candidate process, output, outcome, balancing, equity, and experience measures from a program aim statement.
- Use AI to convert broad goals into SMART measures, operational definitions, numerators, denominators, data sources, reporting frequency, and accountable owners.
- Use agentic AI to pull approved data from dashboards, registries, logs, surveys, or workplan trackers and prepare routine performance summaries for staff review.
- Generate improvement hypotheses, Plan-Do-Study-Act ideas, root-cause prompts, and meeting agendas based on current performance trends.
- Create action trackers that assign owners, due dates, dependencies, status, barriers, and follow-up items for each improvement activity.
- Monitor dashboards for trends, thresholds, missed deadlines, equity gaps, data quality issues, and measures that need governance or leadership attention.

Required Safeguards

- Staff should approve all measures before use so the dashboard reflects public health priorities rather than whatever data are easiest to collect.
- AI-generated measures should be checked for feasibility, validity, equity relevance, burden, and unintended incentives.
- Agentic monitoring should only use approved data sources and should not alter official records, contact partners, or escalate externally without human review.
- Dashboards should show definitions, data freshness, limitations, subgroup views, and interpretation notes so users do not over-read noisy or incomplete data.
- Performance summaries should distinguish observed trends from AI-generated explanations or improvement suggestions.

Learning examples: Draft performance measures from an aim statement, Identify leading and lagging indicators, Generate PDSA ideas from dashboard trends, Monitor dashboard thresholds and equity gaps, Create owner-based improvement action trackers

Related plays: 2, 3, 7, 9, 10, 12, 13. Related tools: 7, 8, 9, 13, 23, 35, 40, 41, 43, 44, 46, 47.

Monitoring, Evaluation, Incident Response, and Continuous Improvement

AI-supported systems require ongoing review after launch. Departments need dashboards, incident processes, policy logs, equity monitoring, and clear retirement or pause criteria.

Useful Activities

- Monitor model performance, data quality, subgroup outcomes, drift, incidents, complaints, and staff adoption.
- Track hallucination or misinformation risks for public-facing outputs.
- Maintain policy update and version control logs.
- Evaluate whether the AI system improves public health outcomes, timeliness, quality, equity, or staff workload.

Required Safeguards

- Define thresholds for pausing, modifying, or retiring systems.
- Assign owners for monitoring and incident response.
- Review high-impact systems through governance on a regular cadence.
- Document lessons learned and update tools, policies, training, and public notices.

Learning examples: Hallucination audit for public-facing products, Incident response for unsafe outputs, Policy update process, Monitor subgroup performance after deployment

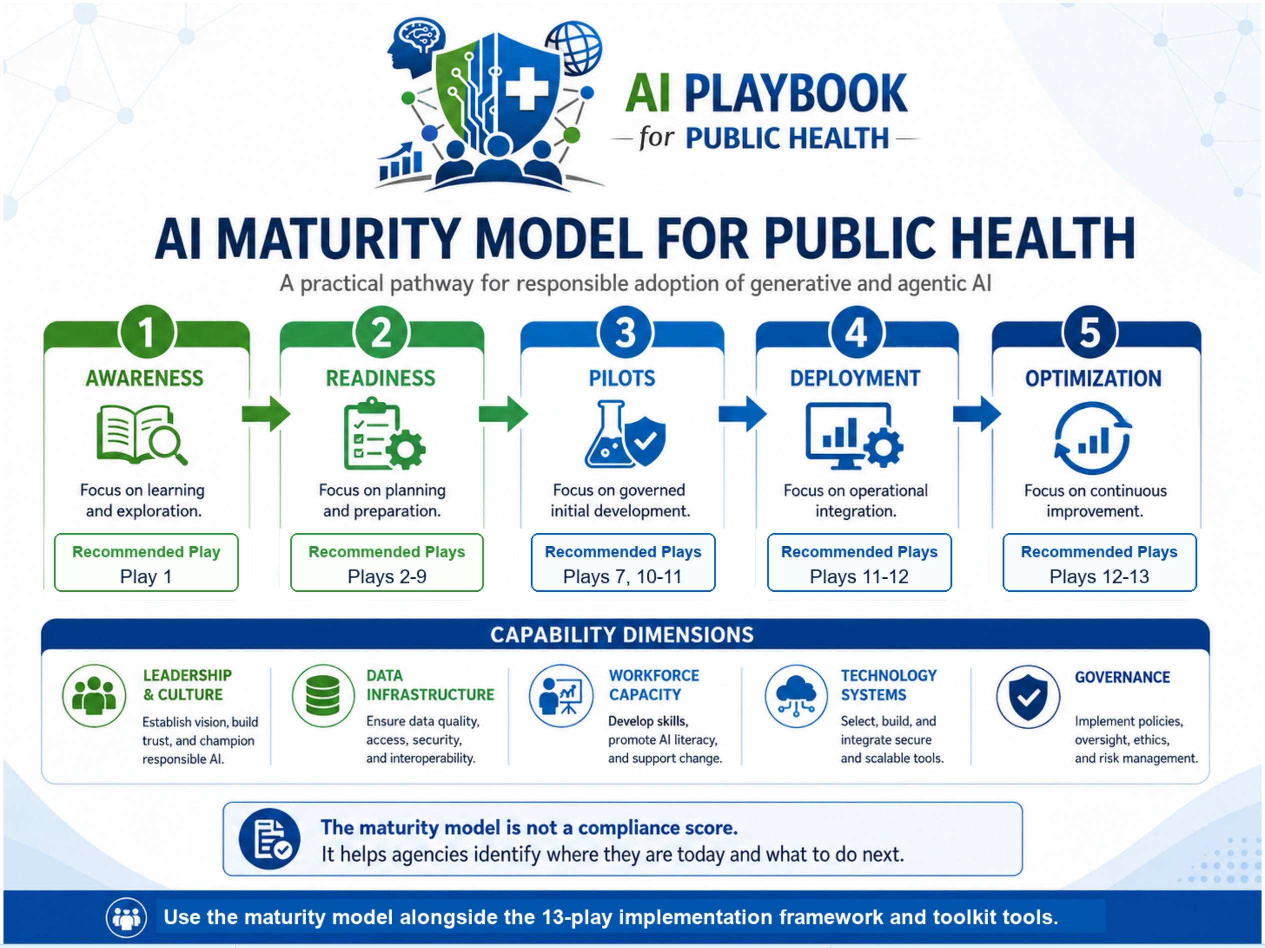
Related plays: 12, 13. Related tools: 14, 17, 18, 40, 41, 43, 44, 45, 46, 47.

Readiness Assessment

The readiness assessment uses a 0-3 question scale and normalizes results to a 100-point scale. Section scores help identify priority gaps; the total score provides an overall readiness estimate.

Domain	Indicators
Leadership and Governance	Leadership has articulated an AI vision aligned with strategic priorities. A clear governance structure exists or is planned. AI ethics and equity principles have been established. Budget authority exists to support AI infrastructure and workforce.
Data Infrastructure	Data systems are integrated or interoperable. Data quality is regularly monitored. Cloud or modern analytics infrastructure is available. Data governance policies exist for access, privacy, security, and use.
Workforce Capacity	Staff with data science, analytics, or AI skills are available. Training programs build AI literacy across the workforce. Job descriptions and career paths support data and AI roles. Academic or technical partnerships provide additional capacity.
Equity and Community Engagement	Equity impact assessment is part of AI planning. Community engagement informs design of community-impacting systems. Local context and applicable community or Tribal data governance expectations are identified and respected. Subgroup performance monitoring is planned before deployment.
Partnerships and Collaboration	Cross-program collaboration supports AI use cases. External partners are identified and governed appropriately. Procurement and vendor pathways are understood. Roles and responsibilities are documented.
Privacy and Security	PHI and sensitive data protections are defined. Security review occurs before AI tools are used. Data use agreements are required for vendors and partners. Incident response and escalation processes exist.
Policy, Procurement, and Public Accountability	AI policy landscape, procurement requirements, public records obligations, and agency restrictions have been reviewed before tools are selected. Vendor and procurement reviews include transparency, data rights, model documentation, auditability, accessibility, security, and exit requirements. Environmental and resource impacts, including compute, energy, water, sustainability, and environmental justice considerations, are assessed for AI projects. Plain-language public notice, human-centered decision rules, and limits on automated decision-making are documented for public-impacting AI uses.

AI Maturity Model



Level	Focus	Recommended Plays
1: Awareness	Learning and exploration	Play 1
2: Readiness	Planning and preparation	Play 2, Play 3, Play 4, Play 5, Play 6, Play 7, Play 8, Play 9
3: Pilots	Governed initial development	Play 7, Play 10, Play 11
4: Deployment	Operational integration	Play 11, Play 13
5: Optimization	Continuous improvement	Play 12, Play 13

13-Play Planning and Implementation Framework

Play 1: Vision & Guardrails

Establish a clear strategic vision for how AI will support the agency's public health mission while defining principles that ensure responsible, ethical, and legally compliant use.

Why it matters: Every AI initiative begins with leadership alignment. Setting guardrails early around equity, human oversight, privacy, and prohibited uses prevents downstream harms and builds institutional trust.

- Convene a 60-90 minute leadership session.
- Agree on non-negotiable guardrails for human oversight, equity, privacy, and prohibited uses.
- Draft a 1-2 page AI Vision and Principles document.
- Align principles with NIST AI RMF 1.0.

- Identify executive sponsorship and update annually.

Outputs

- Draft AI vision statement.
- Draft responsible AI principles.
- Initial guardrails and boundaries.
- List of current known or suspected AI uses.
- Initial strategic priority areas.
- List of unresolved issues requiring legal, privacy, IT, equity, or leadership review.
- Recommended next steps and owners.
- Decision on whether to establish or convene an AI governance committee.

Supporting tools: Tool 1, Tool 2, Tool 3, Tool 4, Tool 5, Tool 6

Play 2: Readiness Assessment

Assess current state across leadership, data infrastructure, workforce, equity, partnerships, privacy, and security so planning reflects organizational reality.

Why it matters: A frank readiness assessment finds gaps while they are still manageable, before a project discovers midstream that data, skills, infrastructure, or governance are insufficient.

- Run a half-day assessment workshop.
- Rate each indicator honestly.
- Identify 5-8 high-priority gaps.
- Assign owners and timelines.
- Brief leadership and the governance sponsors on constraints and next steps.

Outputs

- Completed AI Readiness Self-Assessment.
- Total readiness score on the 100-point scale, with raw score retained for transparency.
- Section scores for each readiness domain: leadership and governance, data infrastructure, workforce capacity, equity and community engagement, partnerships and collaboration, and privacy and security.
- Evidence notes, assumptions, and disagreements used to support each section score.
- Readiness stage interpretation based on the total score.
- Priority weak domains that should shape the next steps, even if the total score is high.
- Gap register identifying the gap, domain, section score, blocking or parallel status, risk level, owner, timeline, related play, and related tool.
- List of blocking gaps that must be addressed before pilots, procurement, deployment, or scaling can move forward.
- List of parallel gaps that can be addressed while planning continues.
- Recommended 2-4 plays to complete next, based on total score and lowest-scoring domains.
- Play prioritization plan showing which plays should be completed first, which can proceed in parallel, and which should wait.
- Recommended supporting tools for each prioritized play.
- 30-, 60-, and 90-day action plan with owners and due dates.
- Leadership or governance briefing summary, including constraints, risks, decisions needed, and recommended next steps.
- Decision list identifying any current or proposed AI activities that should be paused, delayed, revised, or routed to governance review.
- STLT AI policy landscape scan identifying legal, public records, procurement, IT, and agency restrictions.

- Tiered data classification summary showing which data may be used in public, enterprise, in-house, vendor, or restricted AI environments.
- Reassessment plan, including when the department will repeat the assessment and what changes should trigger reassessment.
- Downloadable assessment summary for documentation and governance records.

Supporting tools: Tool 3, Tool 4, Tool 7, Tool 8, Tool 9, Tool 10

Play 3: Establish AI Governance

Create the governance structures, policies, decision-making processes, and oversight mechanisms needed to guide responsible AI use before projects begin.

Why it matters: Governance should not be added after AI tools are already in use. It should be established before use case selection, vendor engagement, pilot approval, or deployment. Public health departments need clear authority, consistent review processes, and documented accountability for AI use. Strong governance helps ensure that AI supports the public health mission while protecting privacy, security, equity, scientific integrity, workforce trust, and community confidence.

- Confirm the governance mandate. Define why AI governance is needed, what decisions it will cover, and how it will support the department's AI vision, principles, and guardrails.
- Establish the governance body. Create or designate a cross-functional AI governance committee or board with clear membership, leadership, authority, and decision rights.
- Adopt responsible AI policy expectations. Define department-wide expectations for transparency, privacy and security, equity, human oversight, safety, legal compliance, scientific integrity, and accountability.
- Create the governance charter. Document the governance body's purpose, scope, membership, meeting cadence, decision process, documentation requirements, and escalation procedures.
- Define the AI review pathway. Establish how proposed AI uses will be submitted, screened, classified by risk, reviewed, approved, paused, or rejected.
- Align governance with existing agency processes. Connect AI oversight with data governance, IT governance, cybersecurity, privacy, legal review, procurement, communications, equity review, program leadership, and quality improvement.
- Set minimum documentation requirements. Define what must be documented for each AI use case, including purpose, owner, users, data involved, risk level, human review process, limitations, approval status, and review schedule.
- Require appropriate review before approval. Ensure that privacy, security, equity, data quality, legal, vendor, and model validation reviews occur when required based on the risk and nature of the AI use case.
- Create an AI system inventory. Maintain a central record of approved AI systems and AI-supported workflows, including owners, purpose, risk level, approval date, monitoring schedule, and review status.
- Establish ongoing oversight. Define how approved AI uses will be monitored, how incidents will be handled, how systems can be paused or retired, and how policies will be updated over time.

Outputs

- AI governance committee or board.
- Responsible AI policy.
- Governance charter.
- Risk-tiered AI review pathway.
- AI use case intake and approval process.
- Documentation standards.
- AI system inventory.
- Tiered data classification rules for AI use.
- Plain-language AI public notice process.
- STLT procurement and vendor review requirements.
- AI-specific incident categories, including hallucinations, bias concerns, unsafe automation, misinformation, privacy exposure, and model drift.

- Monitoring and incident response process.
- Policy update process.

Supporting tools: Tool 2, Tool 3, Tool 4, Tool 10, Tool 11, Tool 12, Tool 13, Tool 14, Tool 15, Tool 16, Tool 17, Tool 18, Tool 19

Play 4: Engage Stakeholders

Build a structured stakeholder engagement framework for internal and external input throughout planning, implementation, and evaluation.

Why it matters: Stakeholder and community engagement is an early planning activity, not a late deployment activity. It should begin after the department has a basic readiness picture and governance pathway, but before formal use case selection, roadmap design, funding commitments, pilots, procurement, or deployment. AI systems built without genuine community and workforce input often fail in deployment even when they perform well technically.

- Map stakeholder groups and assign engagement leads.
- Form advisory groups for community-impacting use cases.
- Share plain-language explanations of what AI will and will not do.
- Hold feedback sessions before use case selection and before design is finalized.
- Document what changed based on feedback.
- Route unresolved stakeholder concerns to the AI governance committee before pilots or procurement move forward.

Outputs

- Stakeholder map.
- Internal stakeholder list.
- External stakeholder list.
- RACI or role clarification matrix.
- Community and partner engagement plan.
- Summary of stakeholder concerns and recommendations.
- Equity and trust considerations.
- Missing voices analysis identifying underrepresented groups, language access needs, disability access needs, tribal sovereignty concerns, rural context, and other perspectives not captured in available data.
- Plain-language explanation plan for communities, partner organizations, and affected populations.
- Documentation of how stakeholder input will inform governance, use case prioritization, implementation, monitoring, and evaluation.
- Recommendations for ongoing engagement or advisory structures.

Supporting tools: Tool 11, Tool 19, Tool 20, Tool 21, Tool 22

Play 5: Workforce Development

Develop the workforce capabilities needed to responsibly adopt generative and agentic AI across public health programs.

Why it matters: Training alone is not enough. Peer learning networks, champions, office hours, and communities of practice help staff build confidence and solve practical problems.

- Define role-based AI literacy levels.
- Conduct a training needs assessment.
- Designate AI champions across divisions.
- Create training pathways.
- Assign learning modules by role and set completion standards.
- Establish office hours and communities of practice.

Outputs

- AI workforce competency framework by role.
- Training needs assessment summary.
- Role-based AI literacy expectations.
- Priority staff groups for training and support.
- AI champion roles and selection criteria.
- AI bridge professional role description and capacity plan.
- Training pathway and delivery plan.
- Role-based learning module assignment matrix.
- Role-based training plan with required modules, optional modules, deadlines, completion standards, owners, and verification approach.
- Training completion and competency tracker.
- Community of practice or office-hours plan.
- Measures for tracking AI literacy, adoption readiness, and ongoing support needs.

Supporting tools: Tool 23, Tool 24, Tool 25, Tool 26, Tool 27, Tool 48

Play 6: Plan Change Management

Plan for adoption before deployment by identifying affected workflows, staff concerns, communication needs, training requirements, and support structures.

Why it matters: Change management starts while the project is still being planned. Early planning prevents technically sound AI projects from failing because staff roles, trust, workflow impacts, or communication needs were addressed too late.

- Draft a change management plan before pilot launch.
- Identify affected staff groups and workflow impacts.
- Prepare staff communication and FAQ materials.
- Surface resistance, privacy, equity, and usability concerns.
- Define training, support, and feedback channels.

Outputs

- Change management plan.
- Affected workflow and staff impact summary.
- Stakeholder and staff communication plan.
- Staff communication and FAQ materials.
- Training and support plan for affected users.
- Resistance, barrier, and concern inventory.
- Mitigation actions with owners and due dates.
- Feedback and escalation pathway for concerns raised before deployment.

Supporting tools: Tool 23, Tool 28, Tool 29

Play 7: Prioritize Use Cases

Identify and prioritize AI use cases aligned with public health priorities and readiness, selecting a small number most likely to deliver meaningful impact.

Why it matters: Disciplined prioritization keeps resources focused and ensures every project is reviewed for equity risk, data feasibility, legal compliance, change readiness, and alignment with the AI vision.

- Invite program areas to submit candidate use cases.

- Score impact, feasibility, equity, scalability, sustainability, and data availability.
- Apply the Use Case Fit Test.
- Select 1-3 initial pilots.
- Submit selected use cases for governance approval.

Outputs

- AI use case intake list.
- Initial use case descriptions and business/public health rationale.
- Use case scoring and prioritization matrix.
- Equity, privacy, legal, data, and operational risk flags.
- Human-centered decision rule for each prioritized use case.
- Missing voices analysis for high-impact or community-facing use cases.
- Environmental and resource impact review for compute-intensive, vendor-hosted, or large-scale AI uses.
- Data readiness and feasibility notes.
- Governance-approved use case portfolio.
- Pilot recommendations, sequencing, and conditions.
- Deferred or rejected use cases with rationale.

Supporting tools: Tool 3, Tool 5, Tool 6, Tool 8, Tool 13, Tool 20, Tool 22, Tool 30

Play 8: Develop a Funding Strategy

Build a coordinated multi-year funding strategy that maps approved use cases and implementation needs to funding sources, staffing, change management, procurement, infrastructure, monitoring, evaluation, and sustainment investments.

Why it matters: AI projects often fail when resources are misaligned: funded for one phase but not the next, approved without dedicated staff, missing change supports, or dependent on missing infrastructure. Funding strategy should shape the roadmap before the agency commits to sequencing.

- Map each approved or proposed project to potential funding sources.
- Document FTE, contractor, infrastructure, training, evaluation, procurement, and sustainment costs.
- Update DMI, PHIG, and other grant work plans.
- Confirm funding assumptions before roadmap sequencing is finalized.
- Review resource allocation quarterly.

Outputs

- Funding strategy for the AI roadmap.
- Funding source inventory and eligibility notes.
- Grant alignment table for approved or proposed AI activities.
- Pilot, scale, sustainment, staffing, infrastructure, training, and evaluation cost estimates.
- Total cost of ownership and sustainability analysis.
- Procurement and vendor cost assumptions.
- Sustainment plan beyond pilot funding.
- Funding gaps, contingencies, and decision points.
- Leadership or governance briefing on resource needs.

Supporting tools: Tool 5, Tool 31, Tool 32, Tool 33, Tool 34

Play 9: Design the Roadmap

Translate vision, governance, readiness, change planning, funding assumptions, and approved use cases into a sequenced multi-year roadmap.

Why it matters: A roadmap prevents isolated pilots and makes shared infrastructure, governance reviews, staffing, funding, change supports, and dependencies visible before implementation starts.

- Complete a project charter for each approved use case.
- Place projects on the implementation timeline.
- Identify shared dependencies and funding assumptions.
- Present the roadmap to governance for approval.
- Publish a one-page roadmap summary and update every six months.

Outputs

- Governance-approved AI roadmap.
- Sequenced implementation timeline.
- Project charters for approved use cases.
- Shared dependencies across data, technology, governance, workforce, funding, and procurement.
- Milestones, decision gates, and review points.
- Resource, staffing, vendor, funding, and infrastructure assumptions.
- Roadmap dependency and risk register.
- Risk register for roadmap delivery.
- Roadmap update cadence and ownership.

Supporting tools: Tool 34, Tool 35, Tool 36

Play 10: Execute Change Management

Execute the change management plan during pilot and deployment by communicating clearly, training staff, supporting adoption, and tracking concerns through governance.

Why it matters: Deployment is where planned change support becomes real. Staff need timely communication, hands-on support, feedback channels, clear escalation paths, and a visible training tracker as workflows change.

- Launch staff communications and FAQs.
- Deliver role-based training and office hours.
- Track completion of required learning modules by role.
- Support change champions with protected time.
- Track staff concerns and barriers.
- Report adoption risks and feedback through governance.

Outputs

- Executed staff communication plan.
- Final staff FAQ and deployment messages.
- Role-based training completion summary.
- Learning module completion status by role or staff group.
- List of staff groups needing follow-up coaching, office hours, champion support, refresher training, or supervisor verification.
- Office hours, support, and change champion activation plan.
- Adoption support log.

- Change management execution dashboard.
- Staff concerns, barriers, and feedback tracker.
- Escalation record for issues routed to governance or leadership.
- Change adoption measures and lessons learned.

Supporting tools: Tool 19, Tool 23, Tool 25, Tool 28, Tool 29, Tool 37, Tool 48

Play 11: Build and Deploy AI Solutions

Develop, test, pilot, deploy, and scale AI solutions that support governance-approved use cases and comply with governance, equity, change management, and validation requirements.

Why it matters: Pilots are for learning. Keep them small, time-bounded, and limited in scope. Scaling is not automatic; it requires explicit governance approval and evidence that adoption support is working.

- Confirm technical infrastructure and data pipelines.
- Test with synthetic or de-identified data.
- Run pilots for 90-120 days.
- Evaluate using pilot dashboard and scale decision framework.
- Submit scale-up plans to governance before expansion.

Outputs

- AI pilot or deployment plan.
- Technical infrastructure readiness checklist.
- Data quality and data use documentation.
- Vendor evaluation record, where applicable.
- STLT AI procurement and vendor addendum, where applicable.
- Decision record for whether public, enterprise, in-house, RAG, or vendor-hosted AI architecture is appropriate for the data tier and use case.
- Model testing and validation protocol.
- Equity and subgroup testing results.
- Incident response and escalation plan.
- Deployment readiness and go-live checklist.
- Pilot evaluation dashboard.
- Pilot-to-scale decision recommendation.
- Governance decision record for deploy, revise, pause, scale, or retire.

Supporting tools: Tool 3, Tool 6, Tool 8, Tool 9, Tool 10, Tool 14, Tool 15, Tool 16, Tool 31, Tool 33, Tool 34, Tool 35, Tool 36, Tool 38, Tool 39, Tool 40, Tool 41, Tool 42

Play 12: Execute Governance and Oversight

Sustain responsible AI through regular governance reviews, policy maintenance, compliance auditing, incident review, version control, and named accountability for every deployed or retired AI system.

Why it matters: Governance intensifies after go-live. The agency must continually ask whether each AI system is still doing what it was intended to do, whether documentation and approvals remain current, whether incidents or monitoring findings require action, and whether the system should continue, change, pause, or be retired.

- Hold governance reviews monthly during pilots and quarterly in sustainment, using a standard agenda and decision log.
- Review performance monitoring, equity monitoring, incident reports, user feedback, audit findings, and continuous improvement actions.

- Confirm that each AI system has a named operational owner, current documentation, approved data use agreements, validation records, and escalation contacts.
- Conduct annual compliance audits and targeted reviews after incidents, model changes, vendor changes, or major policy updates.
- Update policies, charters, review procedures, approval conditions, and version-control logs when guidance, technology, risks, or agency priorities change.
- Document governance decisions, required corrective actions, owners, deadlines, and evidence of closure.
- Define sunset criteria and retire or replace AI systems that no longer meet performance, equity, safety, legal, operational, or public trust standards.

Outputs

- Governance review agenda and decision log.
- AI system inventory updates.
- Incident response records and corrective-action tracking.
- Plain-language public notice updates and public concern review records.
- Human-centered decision rule compliance review.
- Compliance audit checklist and findings.
- Policy update and version control log.
- Review of data use agreements, validation records, and approval conditions.
- Governance decisions to continue, revise, pause, retire, or replace AI systems.
- Documented owners, deadlines, and evidence of closure for required actions.

Supporting tools: Tool 6, Tool 13, Tool 16, Tool 17, Tool 18, Tool 19, Tool 36, Tool 43, Tool 44, Tool 45

Play 13: Monitor and Evaluate AI Systems

Assess whether deployed AI systems perform as expected and deliver measurable public health value with continuous attention to equity and adoption.

Why it matters: Deployment without monitoring is how AI systems cause harm quietly. Monitoring is a core governance function, not an administrative afterthought.

- Monitor reliability monthly and report quarterly.
- Track subgroup performance continuously.
- Conduct hallucination and fabrication audits.
- Evaluate public health outcomes.
- Use a continuous improvement log to track actions.

Outputs

- Performance monitoring dashboard.
- Equity and disparity impact monitoring results.
- Reliability, data quality, model drift, and incident trend summaries.
- Hallucination, misinformation, automation bias, and unsafe output trend review.
- Community complaint, correction, and feedback trend review.
- Missing voices and subgroup performance monitoring results.
- User feedback and adoption findings.
- Outcome evaluation report.
- Continuous improvement log.
- Recommendations to improve, scale, revise, pause, retire, or replace AI systems.

- Monitoring cadence and next review schedule.

Supporting tools: Tool 6, Tool 22, Tool 43, Tool 44, Tool 46, Tool 47

Toolkit Inventory

Tool	Title	Plays	Purpose
1	AI Vision & Principles Workshop	1	Facilitate leadership alignment on mission, principles, priority problems, and prohibitions.
2	Responsible AI Policy Template	1, 3	Create binding commitments on transparency, privacy, equity, human oversight, and legal compliance.
3	Equity Impact Assessment Checklist	1, 2, 3, 7, 11	Evaluate potential disparate impacts and required mitigations.
4	AI Policy Landscape Scan	1, 2, 3	Document state, local, tribal, territorial, agency, and program-specific laws, policies, public records rules, procurement limits, and IT restrictions that shape AI use.
5	AI Environmental and Resource Impact Review	1, 7, 8	Assess energy, water, e-waste, data center, procurement, and environmental justice considerations before approving AI use cases or vendors.
6	Human-Centered AI Decision Rule Checklist	1, 7, 11, 12, 13	Document which decisions AI may support, summarize, prioritize, or draft and which decisions must remain with named human owners.
7	AI Readiness Self-Assessment	2	Assess capacity across leadership, data, workforce, equity, partnerships, privacy, and security.
8	AI Project Charter	2, 7, 11	Define project scope, success criteria, data needs, safeguards, timeline, budget, and risks.
9	Data Quality Assessment Checklist	2, 11	Assess completeness, timeliness, accuracy, interoperability, and data governance.
10	Tiered Data Classification for AI Use	2, 3, 11	Classify data as public, internal, confidential, or highly restricted and define which AI environments may use each tier.
11	AI Advisory Committee Charter	3, 4	Define advisory committee membership, cadence, responsibilities, and decision flow.
12	AI Governance Board Charter	3	Establish authority, membership, voting, meeting cadence, and oversight responsibilities.
13	AI Use Case Intake, Risk Review, and System Inventory Template	3, 7, 12	Capture proposed AI uses, classify risk, document review status, and maintain the approved AI system inventory.
14	Model Validation & Documentation Standards	3, 11	Set documentation, validation, testing, and evidence standards before deployment.
15	Data Use Agreement (AI) Template	3, 11	Document AI-specific data-sharing, privacy, security, and sovereignty requirements.
16	Incident Response Checklist	3, 11, 12	Prepare escalation, containment, documentation, and corrective-action steps.
17	Compliance Audit Checklist	3, 12	Audit compliance with policies, legal requirements, documentation, and monitoring obligations.
18	Policy Update & Version Control Log	3, 12	Track AI governance policy changes, approvals, effective dates, and owners.

Tool	Title	Plays	Purpose
19	Plain-Language AI Public Notice Template	3, 4, 10, 12	Explain what AI is being used, why, what data are involved, what humans decide, known limits, and how people can raise concerns.
20	Community Engagement Planning	4, 7	Plan inclusive engagement, feedback loops, accessibility, and community accountability.
21	Stakeholder Mapping & RACI	4	Map stakeholders, roles, responsibilities, and engagement owners.
22	Missing Voices Analysis	4, 7, 13	Identify groups whose data, perspectives, language needs, access barriers, or community context may be missing from AI design, validation, or monitoring.
23	Change Management Plan	5, 6, 10	Plan communications, training, workflow changes, adoption support, and deployment execution.
24	AI Workforce Competency Framework	5	Define AI literacy, proficient, and practitioner capabilities by role.
25	AI Champion Designation & Role Guide	5, 10	Select and support AI champions who help peers adopt responsible practices.
26	Training Needs Assessment Checklist	5	Identify role-based training gaps before designing training pathways.
27	AI Bridge Professional Role Guide	5	Define the cross-functional role that translates between program staff, IT, leadership, legal, vendors, community partners, and data teams.
28	Staff Communication & FAQ Template	6, 10	Prepare transparent staff communications and answers to expected questions.
29	Resistance & Barrier Identification	6, 10	Surface operational, cultural, and trust barriers before launch and during deployment.
30	Use Case Scoring & Prioritization Matrix	7	Score use cases for impact, feasibility, equity, scalability, sustainability, and risk.
31	Vendor Evaluation Checklist	8, 11	Evaluate vendors for privacy, security, transparency, validation, support, and cost.
32	Funding Strategy Checklists	8	Align projects with funding sources, grant work plans, sustainment, and contingency plans.
33	STLT AI Procurement and Vendor Addendum	8, 11	Add public-sector AI requirements for data residency, audit rights, model training restrictions, cybersecurity, accessibility, public records, exit terms, and subcontractors.
34	AI Total Cost of Ownership and Sustainability Worksheet	8, 9, 11	Estimate pilot, scale, sustainment, staffing, infrastructure, vendor, procurement, training, monitoring, and evaluation costs.
35	AI Implementation Timeline	9, 11	Sequence milestones, dependencies, approvals, pilot phases, and scale decisions.
36	Roadmap Dependency and Risk Register	9, 11, 12	Track roadmap dependencies, governance gates, risks, owners, due dates, mitigation actions, and escalation decisions.
37	Change Management Execution Dashboard	10	Track training completion, communication delivery, adoption support, staff feedback, barriers, escalations, and deployment readiness during change execution.
38	Technical Infrastructure Readiness Checklist	11	Confirm environments, access controls, integrations, monitoring, and secure deployment paths.

Tool	Title	Plays	Purpose
39	AI Model Testing & Validation Protocol	11	Define test design, performance thresholds, human review, and validation evidence.
40	AI Pilot Evaluation Dashboard	11	Track pilot performance, adoption, equity, incidents, and go/no-go evidence.
41	Pilot-to-Scale Decision Framework	11	Support governance decisions on scale, revise, pause, or retire after pilot evaluation.
42	AI Deployment Readiness and Go-Live Checklist	11	Confirm governance approval, validation, security, data readiness, staff training, communications, support, rollback plans, and go-live decision criteria before deployment.
43	Performance Monitoring Dashboard	12, 13	Track reliability, performance, equity, incident trends, and improvement actions.
44	Continuous Improvement Log	12, 13	Track improvement actions, owners, timelines, evidence, and closure.
45	AI Governance Review Meeting Template	12	Structure governance agendas, decisions, escalations, and action tracking.
46	Equity & Disparity Impact Monitoring	13	Monitor subgroup performance and disparate impacts after deployment.
47	AI Evaluation & Outcomes Reporting	13	Report public health outcomes, operational effects, and lessons learned.
48	Role-Based Training Plan and Completion Tracker	5, 10	Assign learning modules by role, set completion standards, track deadlines, verify completion, and identify follow-up support needs.

Case Studies and Promising Practices

California Health Care Facility Inspection Documentation

Challenge: Health care facility inspectors must document findings clearly and connect observations to applicable laws, regulations, and requirements. That work can be time-consuming and difficult to standardize across inspection teams.

AI solution: California's GenAI program identifies this as a CDPH pilot testing generative AI support for inspection report drafting and regulatory linkage.

Workflow impact: Inspectors would continue performing inspections and exercising professional judgment, while generative AI helps draft or organize report language and identify potentially relevant regulatory citations for human review.

Outcomes: The expected benefit is reduced documentation burden and faster linkage between inspection findings and applicable requirements. Because this is described as a pilot, the page should not present final outcome claims until public evaluation results are available.

California Language Access Translation Support

Challenge: Residents with limited English proficiency need timely, understandable, and culturally appropriate access to health and social services information. Traditional translation workflows can be slow, inconsistent, or difficult to scale across programs.

AI solution: California's GenAI language access work explores generative AI to improve translation speed, consistency, efficiency, and cultural awareness, with human review and editing.

Workflow impact: Generative AI can create first-draft translations or language-access materials that qualified staff review, edit, and approve before use. The workflow shifts translation from a fully manual first draft to an AI-assisted draft plus human quality review.

Outcomes: The expected benefit is faster, more consistent translation support for public-facing health and social services materials. Public-facing use should still depend on human review, quality assurance, and language access governance.

Kansas Local Health Department AI Roadmap and GenAI Tools

Challenge: Local health departments need practical guidance on whether and how to use generative AI for routine operations, communications, grant writing, health literacy, and content review without creating privacy, quality, or trust risks.

AI solution: The Kansas Public Health Collaborative created AI guidance and roadmap resources for local health departments, including practical GenAI tools and FAQs for communications, grant writing, health literacy, campaign planning, and content review.

Workflow impact: Local health departments can use the roadmap and tools to move from informal experimentation toward structured, governed use of GenAI in communications and administrative workflows.

Outcomes: The primary documented outcome is the availability of a roadmap, tools, and community-of-practice support for Kansas local health departments. The page should not claim implementation impact until local deployment outcomes are publicly documented.

Egyptian Health Department Agentic Operations Platform

Challenge: A regional public health and human services agency needed to coordinate screenings, referrals, client information, reporting, and program workflows across public health and human services operations.

AI solution: The agency implemented Creatio's CRM platform, described by the vendor as supporting automation and AI-ready operations for care coordination, reporting, eligibility validation, client portals, and population health workflows.

Workflow impact: The platform centralizes client and program workflows so staff can coordinate referrals, manage screenings, generate reports, and prepare for future AI-assisted risk identification or forecasting within a unified operational system.

Outcomes: The vendor case study reports reduced manual data entry, faster reporting, and improved operational coordination. Because the main source is a vendor case study, the website presents this as an operational example rather than a validated best practice.

Outbreak Early Warning and Investigation Support

Challenge: A state health department needed to identify potential infectious disease outbreaks more quickly using multiple data streams, including syndromic surveillance, laboratory reports, and emergency department chief complaints. Manual review processes delayed detection and limited the department's ability to identify geographic clusters early enough for timely investigation.

Solution: The department used machine learning to detect unusual patterns and potential geographic clusters that could indicate an emerging outbreak. The system was designed as a human-in-the-loop decision-support tool. It flagged potential clusters for review by epidemiologists, but it did not automatically issue public notifications, initiate enforcement actions, or make final outbreak determinations.

Data: The system used syndromic surveillance data, laboratory reports, emergency department chief complaints, geographic data, and related public health surveillance records. During readiness assessment, the department identified a 48-hour lag in syndromic surveillance reporting and addressed that gap through a data pipeline improvement before launching the pilot.

Workflow: The AI system shifted the department from primarily manual review toward earlier automated signal detection followed by human investigation. Epidemiologists received prioritized cluster alerts instead of manually scanning multiple data sources. This allowed staff to focus time on validating signals, contacting local partners, and determining whether additional public health action was needed.

Outcomes: In the first year, the system detected three potential outbreak clusters that were later confirmed through investigation. Average detection time decreased by 4 days compared with the department's previous manual review process. The system improved situational awareness while preserving human judgment in all investigation and response decisions.

Ethics: The department established clear guardrails before implementation. The AI system could only flag geographic clusters for human review and could not trigger automatic public alerts or enforcement actions. Community engagement surfaced concerns from immigrant communities about data use, leading the department to add explicit language to its AI principles prohibiting the use or sharing of individual-level data for law enforcement purposes.

Agentic: The system could evolve into an agentic AI workflow by allowing the AI to take defined, bounded actions after a potential cluster is detected. For example, the AI could automatically gather supporting laboratory, syndromic, emergency department, and geographic data; generate a preliminary investigation brief; compare the signal against historical baselines; create a draft case investigation ticket; recommend next investigation steps; and route the alert to the appropriate epidemiology team based on geography, disease category, or staffing availability. The AI could also monitor incoming data after the alert is created and update the investigation brief as new information becomes available. Human epidemiologists would still review the evidence, validate the signal, approve any external communications, and decide whether public health action is warranted. Additional safeguards would be needed, including role-based permissions, audit logs for every AI-initiated action, human approval before external communication, limits on what data the AI can access, and explicit prohibitions on enforcement or law enforcement uses.

Tribal Chronic Disease Risk Stratification for Care Management

Challenge: A tribal health department needed a more systematic way to identify tribal members at elevated risk for type 2 diabetes complications and prioritize care management outreach. Existing outreach processes relied heavily on manual review and did not consistently identify the members who could benefit most from early intervention.

Solution: The department used AI-assisted risk stratification to identify tribal members at higher risk for diabetes-related complications. The tool supported, but did not replace, clinical and care management judgment. Tribal health staff used the risk scores to prioritize outreach, review individual needs, and determine appropriate follow-up.

Data: The model used tribal health department data, including clinical records, diabetes-related indicators, care management records, utilization history, demographic information, and other locally governed data sources approved for use by tribal leadership. All data remained under tribal jurisdiction.

Workflow: The AI tool helped care management staff move from broad, manual outreach lists to more targeted outreach based on risk level. Staff could prioritize members most likely to benefit from follow-up, review each case, and coordinate outreach through existing tribal health workflows. The system supported earlier intervention while keeping decisions and outreach activities within the tribal health department.

Outcomes: Care management outreach rates increased by 35% among high-risk members. The project improved the department's ability to identify and engage members at elevated risk for diabetes complications. The governance model was also shared with other tribal health departments as an example of how AI oversight can be adapted to a tribal sovereignty context.

Ethics: The department's AI principles explicitly recognized tribal sovereignty and data governance. All data remained under tribal jurisdiction, and no data could be shared with state or federal systems without explicit tribal council approval. AI tools were operated only by tribal health staff. Governance oversight was assigned to the tribal council, with reporting to the tribal health board rather than to a state health department.

Agentic: The system could evolve into an agentic AI workflow by allowing the AI to support defined care management tasks under tribal health staff supervision. For example, the AI could identify members whose risk scores have increased, prepare a prioritized outreach list, generate a draft care management summary for staff review, suggest culturally appropriate outreach scripts, recommend follow-up intervals, and create internal task reminders for care managers. The AI could also monitor whether follow-up visits, lab tests, referrals, or education sessions were completed and flag cases that require additional attention. Tribal health staff would retain authority over outreach decisions, care planning, member communication, and any referrals or clinical actions. Additional controls would be needed, including tribal approval of allowable AI actions, audit logs, human review before member contact, culturally grounded communication standards, opt-out processes where appropriate, and strict limits preventing the AI from sharing data outside tribal governance structures.

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Responsible AI and Federal Guidance

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CDC STLT resources. Evolving guidance for state, territorial, local, and tribal public health agencies. https://www.cdc.gov/public-health-gateway/php/index.html

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HIPAA and Business Associate Agreements. Privacy compliance guidance when protected health information may be processed by vendors, cloud systems, or AI tools. https://www.hhs.gov/hipaa/index.html

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NIH / NSF smart health programs. Support for research partnerships, model evaluation, public health AI development, and agentic system research. https://new.nsf.gov/funding/opportunities/smart-health-biomedical-research-era-artificial

SAMHSA technology-based services. Relevant to early identification, intervention support, client communication, and care coordination. https://www.samhsa.gov/grants

EPA environmental justice grants. Relevant to environmental health analysis, GeoAI, and heat risk mapping. https://www.epa.gov/environmentaljustice/environmental-justice-grants-funding-and-technical-assistance

FEMA emergency management grants. Relevant to disaster resilience and agentic outbreak forecasting workflows. https://www.fema.gov/grants

Public Health Peer Networks and Partners

CDC STLT Data Connection. Peer-learning and community-of-practice connection for public health data modernization and AI capacity building. https://www.cdc.gov/data-modernization/php/stlt-data-connection/index.html

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AI Platforms and Model Ecosystem Examples

OpenAI GPT-4 and GPT-4o. Leading large language models available through commercial platforms such as Microsoft Azure OpenAI. https://openai.com/api/

Microsoft Azure OpenAI. Commercially hosted API option that may support enterprise controls and Business Associate Agreement pathways. https://azure.microsoft.com/products/ai-services/openai-service

Anthropic Claude. Large language model family and Deep Research-capable vendor ecosystem. https://www.anthropic.com/claude

Google Gemini and Google Vertex AI. Commercial AI models and hosted AI platforms relevant to public health agency procurement decisions. https://cloud.google.com/vertex-ai

Meta Llama. Open-source model option that agencies may consider for controlled or self-managed infrastructure. https://www.llama.com/

Deep Research tools from Google, OpenAI, and Anthropic. Practical agentic capabilities for literature synthesis, legal scans, policy scans, scenario analysis, and comparative program analysis. https://openai.com/index/introducing-deep-research/

Risk, Validation, and Governance Concepts

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Automation bias. Governance risk that occurs when staff reduce scrutiny after repeated exposure to AI outputs. https://www.nist.gov/itl/ai-risk-management-framework

Model drift and performance monitoring. Sustainment risks requiring ongoing dashboards, audits, and continuous improvement. https://www.nist.gov/itl/ai-risk-management-framework

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In the News

Curated developments in generative and agentic AI that may affect state, territorial, local, and tribal public health departments.

Colorado lawmakers introduce new AI rules for automated decisions

Source: Axios Denver. Date: May 3, 2026. Category: State AI Legislation.

Colorado lawmakers introduced a replacement framework for the state's AI law that would regulate automated decision-making in areas including health care, insurance, employment, housing, education, and financial services.

Why it matters for public health: Public health departments should watch this because state AI rules may affect procurement, vendor documentation, human review, data correction rights, risk review, and the use of AI in eligibility, services, inspections, case management, and health programs.

Original source: https://www.axios.com/local/denver/2026/05/03/colorado-ai-artificial-intellligence-legislation

New York lawmakers consider guardrails for high-risk AI decisions

Source: Times Union. Date: January 15, 2026. Category: State AI Legislation.

New York lawmakers examined proposed AI rules for high-risk decision-making, including requirements related to discrimination prevention, audits, transparency, and human review in sectors such as health care, housing, employment, and public services.

Why it matters for public health: Relevant to health departments because similar state bills could require documented risk controls, bias review, appeal pathways, human oversight, and audit-ready records when AI supports public health decisions or services.

Original source: https://www.timesunion.com/state/article/new-york-lawmakers-eye-new-ai-regulations-21297051.php

New York legislature advances several AI regulation bills

Source: Times Union. Date: June 4, 2026. Category: State AI Legislation.

New York lawmakers moved multiple AI bills near the end of session, including measures focused on AI training data transparency, chatbot risks, workforce impacts, and AI-generated media disclosure.

Why it matters for public health: Public health departments should monitor these types of bills because they may affect staff training, chatbot use, workforce planning, public communications, transparency expectations, and records needed for AI-supported workflows.

Original source: https://www.timesunion.com/capitol/article/legislature-pass-several-ai-regulations-end-22291385.php

Federal AI draft could override some state AI laws

Source: Axios. Date: June 4, 2026. Category: State AI Legislation.

A bipartisan House discussion draft for a national AI framework would preempt some state AI laws while adding provisions related to AI whistleblowers, fraud penalties, AI literacy, education, and research.

Why it matters for public health: This matters for public health departments tracking state AI obligations because federal preemption proposals could change whether departments follow state-specific rules, federal requirements, or both.

Original source: https://www.axios.com/2026/06/04/house-draft-bill-regulate-ai

California announces discounted Claude access for state government

Source: TechRadar. Date: June 30, 2026. Category: Government AI.

California announced a statewide arrangement to make Claude available to government agencies at reduced cost, reinforcing the need for procurement, access controls, approved-use guidance, and workforce training before broad GenAI access expands.

Why it matters for public health: Useful for public health departments thinking about enterprise AI access, member permissions, role-based guidance, and how state-level procurement decisions may affect local implementation.

Original source:
https://www.techradar.com/pro/newsom-strikes-anthropic-deal-to-get-california-government-half-price-claude-ai-access

FDA launches agencywide AI tool

Source: Axios. Date: June 2, 2025. Category: Federal Health AI.

FDA introduced an internal AI tool for agencywide use, showing how health agencies are moving from experimentation toward enterprise AI environments with governance, acceptable-use boundaries, and operational support.

Why it matters for public health: Helpful context for health departments planning approved AI environments, staff guidance, and governance review for internal AI assistants.

Original source: https://www.axios.com/2025/06/02/fda-launches-ai-tool

California inspection report search tool uses AI-assisted public records workflows

Source: San Francisco Chronicle. Date: December 22, 2025. Category: Transparency and Inspection Data.

A newsroom project used AI-supported search and document processing to make California health facility inspection reports easier to find and analyze, illustrating public demand for clearer access to complex regulatory information.

Why it matters for public health: Relevant to public health teams considering AI-supported inspection documentation, regulatory linkage, document search, public transparency, and quality review workflows.

Original source: https://www.sfchronicle.com/about/newsroomnews/article/hospital-tool-california-21244615.php

Opportunities and risks of generative AI across the health information journey

Source: arXiv. Date: May 21, 2026. Category: Evidence and Risk.

This paper examines how generative AI may affect the health information journey, including possible benefits and risks across information creation, interpretation, access, and use.

Why it matters for public health: Supports learning modules on responsible prompting, evidence review, misinformation risk, communication quality, and human review.

Original source: https://arxiv.org/abs/2605.23026

Agentic AI governance and lifecycle management in healthcare

Source: arXiv. Date: January 22, 2026. Category: Agentic AI Governance.

This paper focuses on governance and lifecycle management for agentic AI in healthcare, including oversight needs when systems can plan, act, monitor, and adapt within defined boundaries.

Why it matters for public health: Useful for public health governance, system inventory, monitoring, incident response, role permissions, and organization-level accountability design.

Original source: https://arxiv.org/abs/2601.15630

HIPAA-compliant agentic AI system design in healthcare

Source: arXiv. Date: April 24, 2025. Category: Privacy and Security.

This paper discusses design considerations for agentic AI systems in healthcare environments where privacy, auditability, and compliance controls are central.

Why it matters for public health: Helpful for departments reviewing privacy safeguards, approved environments, audit logs, data minimization, and vendor or architecture requirements.

Original source: https://arxiv.org/abs/2504.17669